

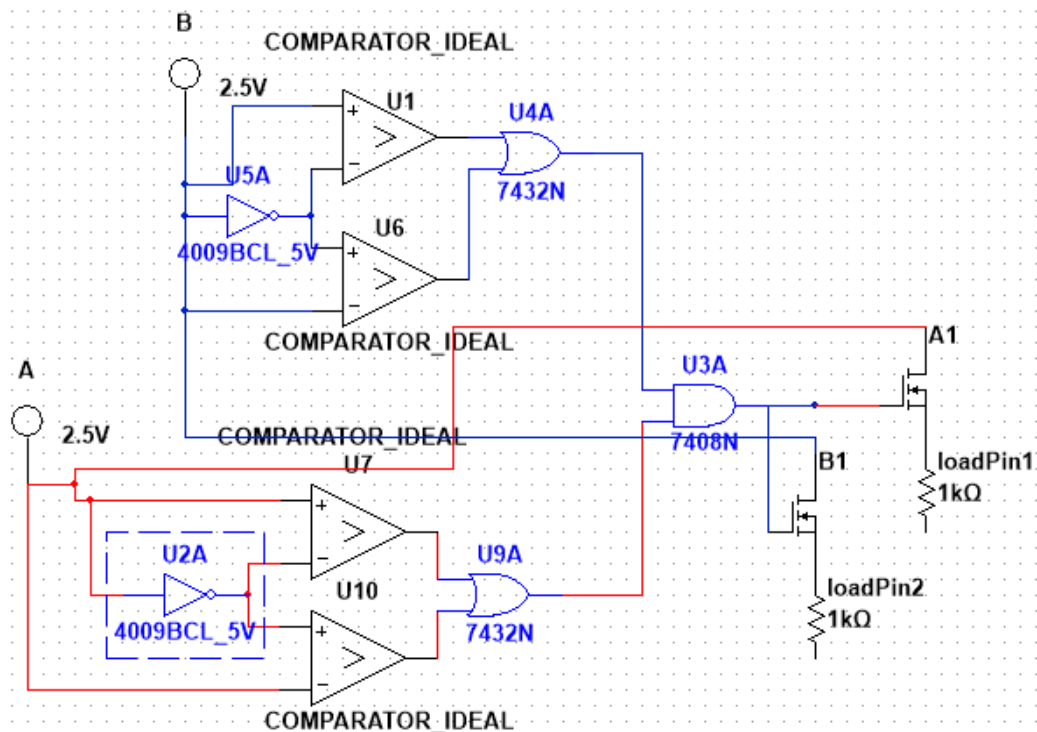
11/5

1. I determined the problem! **When working with devices that require synchronized analog signals input, the usual ways to synchronize signals are costly and complex.**
2. I did some research. What do people usually do before: Length Matching. Problems with that: wires cost a lot in industries, and the longer the wire, the more electromagnetic interference there is. Phase-Locked Loops (PLLs) & Delay-Locked Loops (DLLs): These are complex feedback control circuits that actively manage the clock signal. They are complex, create noise.

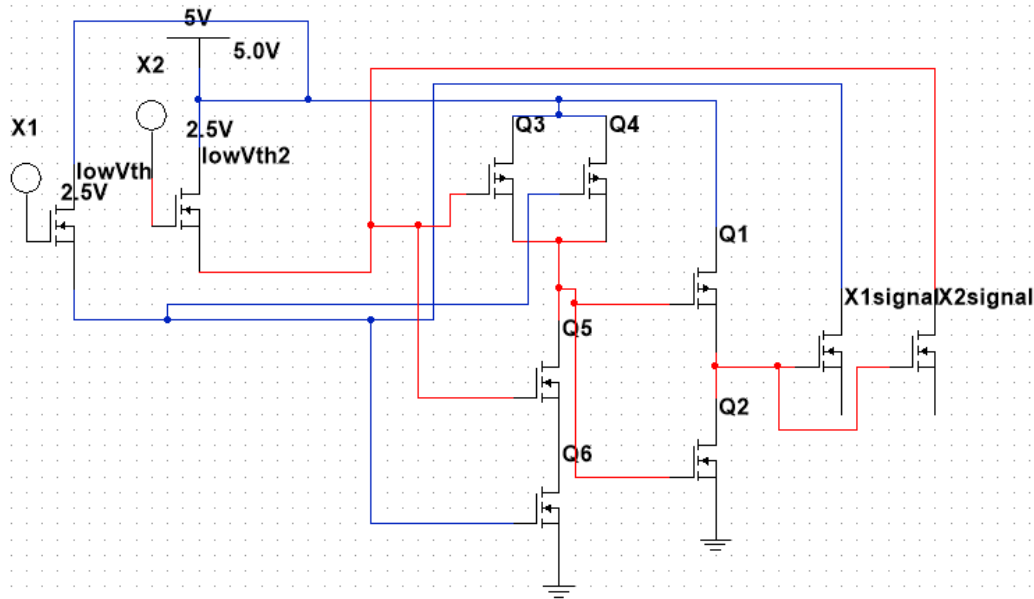
11/10

1. The first draft and the alternative circuit with only mosfet finished drawing.
2. I think if I compare it to itself then it will even detect a negative signal! The logic is simple, if there is an input signal B, the comparator U1 will give a high voltage or U6 will give a high output, then the or gate will check for if one comparator passes a high voltage, which indicates that a signal passes, then, the output will go to the and gate at U3A. The signal input A goes through a similar process as B, and if both or gates output a high voltage, U3A will output a high voltage, which connects to the MOSFETS as switches to control the signals to pass through or not. If U3A outputs a high voltage, then, both the MOSFETs allow A and B to pass through.

This is the first draft:



In the alternative circuit, I'm thinking to use only MOSFETs to build a circuit at 0 energy cost.



11/11

1. My dad looked at the circuits after I completed them by myself. **He asked me to try to make the circuit simple.**
2. **I decided to stay with the circuit built previously because it is a model, and I will just use CMOS logic gates in my circuit to minimize energy loss because the alternative design will take too much effort to build with only MOSFETs.**
3. My dad pointed out a stupid mistake I made. The inverter switches from 1 to 0 or 0 to 1, **he asks me to try to find a way to make the signal actually from + to - or - to +.**

11/12

1. **I fixed the problem. I used an operational amplifier with negative coefficient to time the signal so that the circuit may work.**
2. I'm now thinking, what if I make the circuit more innovative? If I use no MOSFETs, since MOSFET may have very slight timing differences in opening as switches, if I use capacitors? I can utilize capacitors to serve as a switch and replace the logic gates. Since in series, the capacitor's voltage grows gradually determined by the capacitance and the resistance in the circuit, what if I use a capacitor to delay the signal from directly going to the device?
3. My dad asked me, **how am I going to build the circuit?** I said if I use a MCU to open up a MOSFET switch or something, but I realized that is making it more complex, and I still use MOSFETs in the new method, there is no difference between the original plan but making it more complex and costly.

11/13

1. I'm thinking what if I solve the problem with industries that use different **signals like analog v.s. Digital, high voltage v.s. Low voltage needs to be separated.**
2. I realize that It will be **too hard for me to complete with my knowledge.**

11/14

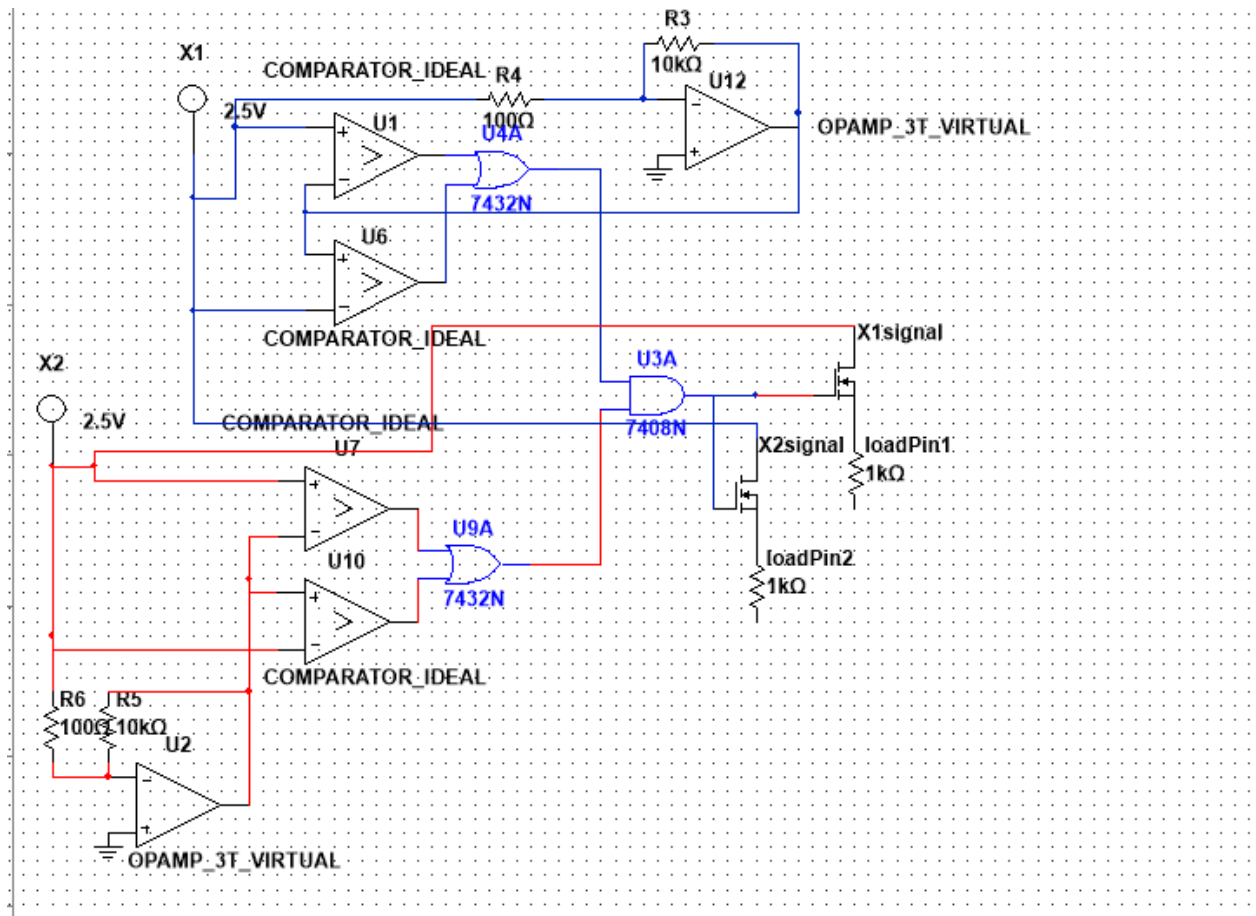
1. As I'm walking on the street, I look at the wind blowing the trees. I'm relating wind to electromagnetic interference to the wind, and the leaf on the trees to the devices. However, the grass on the ground gets affected less. And shorter tree branches get less affected.
2. I'm thinking, since there are electromagnetic interferences, and the longer the wire, the more EMI. **What if I can put the device to a point where there is less EMI? What if something can kind of absorb, or interfere with the magnetic field or just in any way to stop EMI or attract EMI. However, I find that the simplest way in factories to prevent EMI is to place metal walls to separate the devices and wires. So, this idea fails.**

11/17

1. I'm reflecting on my circuit. I'm thinking, what if the signal is so small and the comparator cannot distinguish even if it is comparing to the opposite sign of itself.

11/18

1. I modified the coefficient to $\ast(-100)$ so that even if the signal is small it still works.



11/19

1. **My dad asked me to think about what will happen if my coefficient is too large.**
2. **I'm thinking about my circuit, if I made the coefficient to (-100), that limits the range of the signal accepted or the comparator cannot compare a too large signal.** I need to find an appropriate coefficient and set a range to the signals. I need to research more about signals' voltage ranges. In regard to voltage ranges, what if I have a variable resistor depending on the input voltage sizes? Like a wisdom coefficient that automatically analyzes past input and adjusts the size of the resistor.

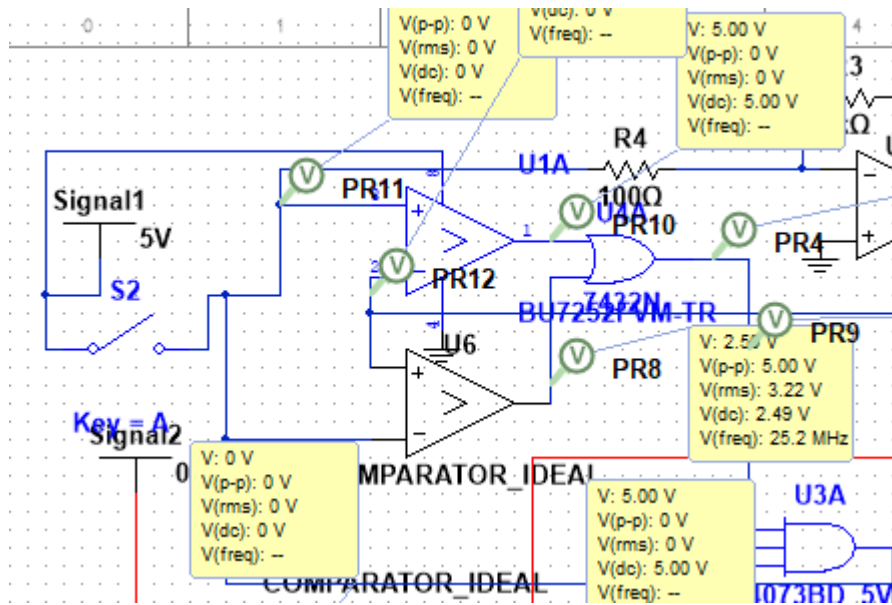
11/20

1. What if I add functions to my design and make it encode the signals and decode it to tell the status, like if it's a high/low signal, or even use other categories. And, if the signal someday suddenly changes, and the code changes, report to a device that it may be not working.
2. In the future, I will think about whether I can add extra components, like if an input signal is not working, use a capacitor's output to simulate the old signal size. I also need pins, 1 to turn on/off the whole device, one to turn on/off the use of the capacitor method. One to turn on the device while preventing the signals from flowing to the device, like intentional delaying. I got this idea from the 74HC148 encoder.
3. If I want to use the capacitor to output similar signals, I'm thinking now to use the attribute of diodes. When in parallel, the circuit's voltage at the parallel will be equal to the voltage drop of the diode. If I use a variety of diodes with the different voltage drop, and test the input signal's voltage, then, when the signal suddenly stops, the specific diode matches its voltage drop will somehow make the capacitor output a similar voltage. And this can also achieve a function that when the device pin only accepts certain signals, and if the signal suddenly changes, use the capacitor to simulate old signals, and then it maintains the old signal until the original signal changes back. I will use the outputs from different comparators, 1 (when the input is a high voltage) I use the original output, 1 (when the input is small) I add an inverter to the output of the comparator.
4. Now I actually want to add the detection to signals that possibly go wrong because I am focusing more on the signal sides, and why not add a function to detect bad signals.
5. When I'm simulating my circuit, I face a problem of the output of one of the 2 comparators for each signal output is always high, which results in the circuit working like all signals come even though the signal does not come. However, I realized that I used the ideal comparator so even slight differences in voltage makes it work. Then, I changed to a realistic component, it's called this (I just found a random one to test) "BA2903YFVM-CTR". However, the output voltage of the comparator grows very slowly, maybe I need to find a correct comparator. I changed to another high speed component, but it works almost like the realistic one, maybe I should lower my coefficient for the operational amplifier. I changed the resistance to the switch, then, there are no voltages passing through the switch when it's not off. But it's still outputting a high voltage. Multisim keeps bumping out notifications as something is wrong. I think maybe it's because of that, I will see what happens when I build the physical circuit.

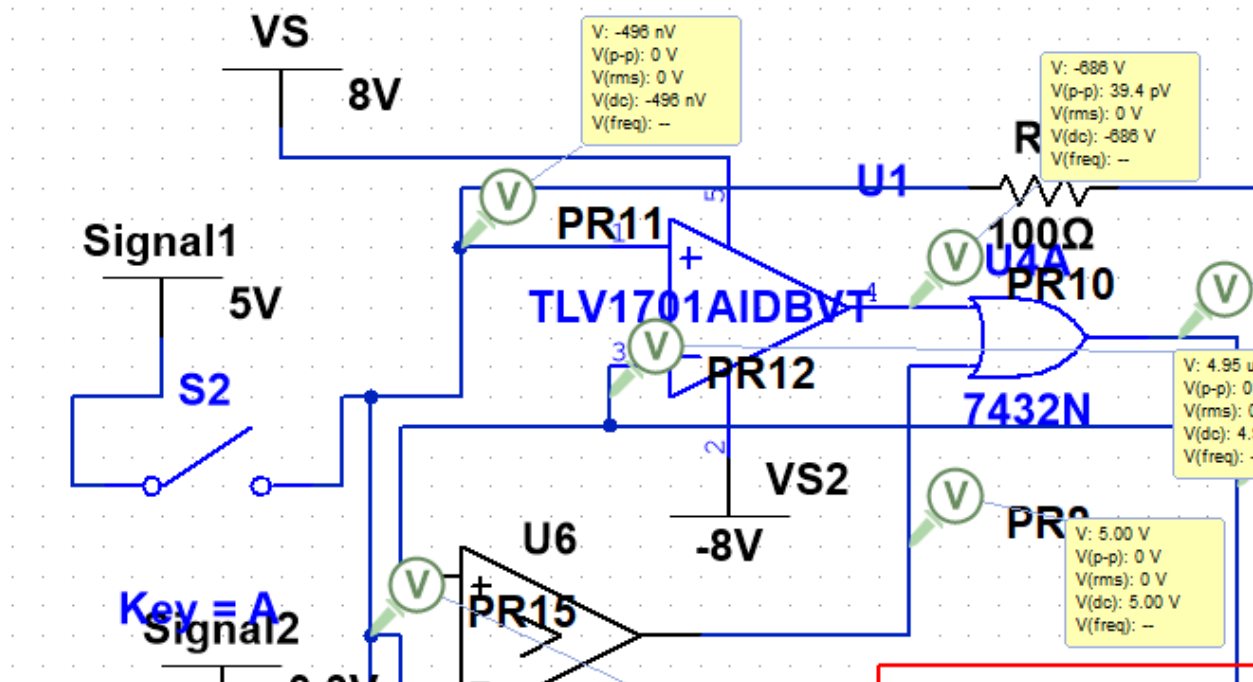
11/22

1. Maybe the comparator issue is because it is assumed as an ideal comparator in multisim and there is a slight offset voltage already.

2. Multisim keeps telling me that something is wrong when simulating, and they cannot fix it.



3. The second comparator in the screenshot's output keeps changing from high to low and to high.
4. I have LM393 which has very little offset voltage.
5. Unfortunately, LM393 does not support minimum input to lower than 0V. I'm buying LM339, it allows negative input min to -0.3V, then, I need to find out a new negative coefficient for the operational amplifier part, and I need to research on usual signal sizes in industries.
6. I just realized I need to change my MOSFET part. If the input voltage is low, or even negative, that will cause problems. I need to change it.
7. Now, I figure out that in the datasheet it says that LM393 allows input from -0.3 V to 36 V, so the first problem is solved.
8. However, I just realized that the absolute minimum is -0.3 V, but it recommends above 0V, so I need to find another one that has recommended input less than 0 V for better performance.
9. I found TLV1821, it has a recommended minimum of -0.2 V! So now I will stay with this comparator. It only has 500 uV offset voltage! So now the issue moves to having a negative supply voltage in the down part of the comparator. I can use an operational amplifier with a negative coefficient to achieve that goal.
10. I found a better comparator, TLV 1701, it allows (VS-)-0.5 minimum input voltage, it is great! I will buy the TLV1704 because it's four channels, and it saves space on my board.



I have no idea why the output is -686 V when I change to TLV1701. Multisim is such an interesting software. I don't trust it now. There is no way that the supply voltages are this little and it still outputs too high.

11/23

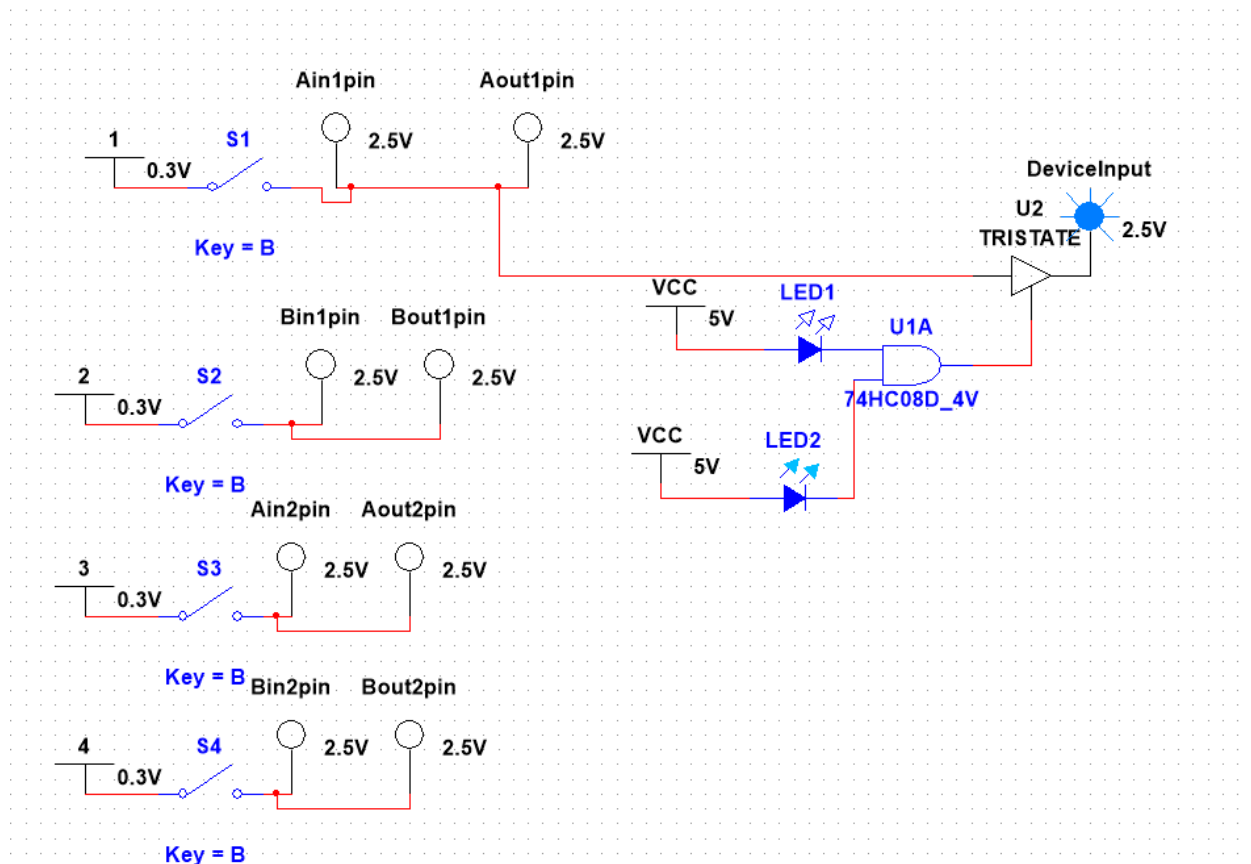
1. I'm trying to solve the problem of if the input is too low and the NMOSFET can't allow it to pass.
2. If I use one n times coefficient op amp before the NMOSFET, and another $1/n$ times coefficient op amp after the NMOSFET. But, there will be differences between the new input to the original input due to lack of precision in the op amp. Then, I realize that differences in resistor size due to lack of problem may also cause problems.
3. If I have a PMOSFET in parallel with the NMOSFET, and when the AND gate passes a high (5V), when the signal is above 5.7 V, the NMOSFET allows the signal to pass through. When the signal is below 5.7 V, the PMOSFET allows the signal to pass through.
4. Then, I figured out that I remembered the wrong condition for MOSFET to work, it does not require the input voltage > gate voltage, so it's fine.
5. However, another thing I need to be aware of is MOSFET's voltage range. I need to find one type of NMOSFET to use and find the data for it. I only have 1 type of NMOSFET 2N7000.
6. I need to change the components to the components I'm using instead of ideal and virtual components, make the upper and lower voltages for the comparators and op amps the same in magnitude.
7. I should make the coefficient just (-1) because of the input tolerance of my comparator. I also forgot to change the OR gates to CMOS gates to save energy.
8. I have these op amps already at home: LM358, LM324, LM380, NE5532, UA741. From the ones I have, only UA741 and NE5532 are designed for dual supply. Overall, NE5532 allows a wider input and V supply ranges, and it has smaller input offset voltage and current in the datasheet. So I will use NE5532.
9. This is the edited version.

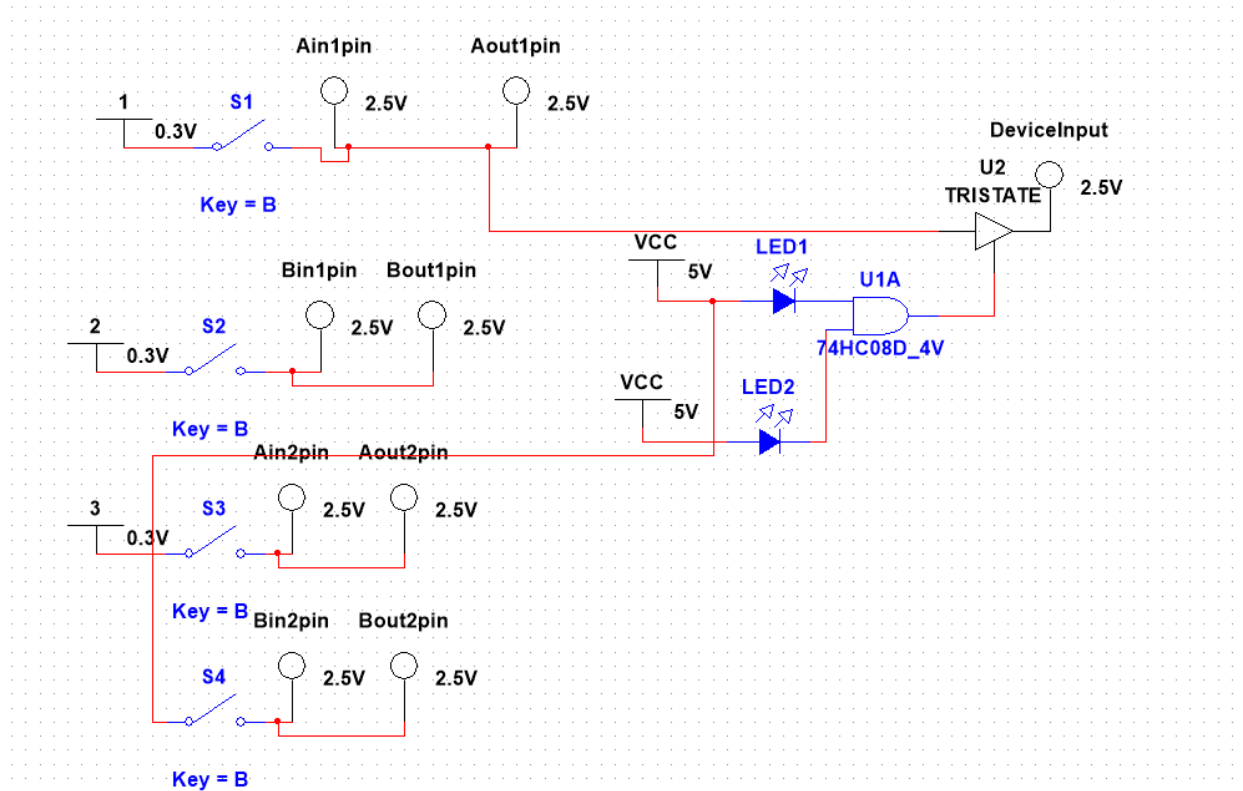
11. **I think I will use a three-state gate with a similar pin with EnableDisable.** I realize that I don't need a three-state gate. **I require the user to input a high voltage to enable the function, so I just need a NMOSFET and that's it.** But, NMOSFETs require consistent input voltage, **maybe I will use a capacitor to store energy and power the NMOSFET.** The energy comes from VCC.

11/24

1. MCU can input a consistent voltage, so there is no need to make it complicated.
2. I also realize that I need to add an inverter to the enable disable pin so that it is automatically enabled unless input a high voltage from an MCU.
3. I also changed the NMOSFETs to the ones I'm going to use. **I'm making the functional part automatically usable because it will be extra work to enable it.** And I can leave an option to close the function if not needed. So I just **added an inverter gate.**
4. **I also updated the resistance for the resistors connected to the op amps to minimize energy loss, because $P=VI$, $I=V/R$, the greater the R , the smaller the I , the smaller the P .**

5. I use 5k1 resistors. I also update the logic gates to the ones I'm using.
6. **I also add a LED to show if it is working or not.**
7. **I'm thinking since my design is just a model of 2 inputs, what if consumers have more inputs like 4?**
8. I will add another pin out for the LED and if it works, the pin has a high voltage there. When the amount of input is even, if the user add an AND gate connecting two LED pins and connect it to a three state gate, when the output of the and gate is 1, all the signals (2*amount of my design) will pass to the device. If the amount of input signals is odd, no problem, connect the leftover signal to input pin A, and connect an LED pin from other parts to the input pin B on the same device with the leftover signal. And connect all LED pins output like I described above for even amount of inputs. I will draw the part out. The user will just add one tristate to one device input pin. **(the screenshot is the demonstration).**



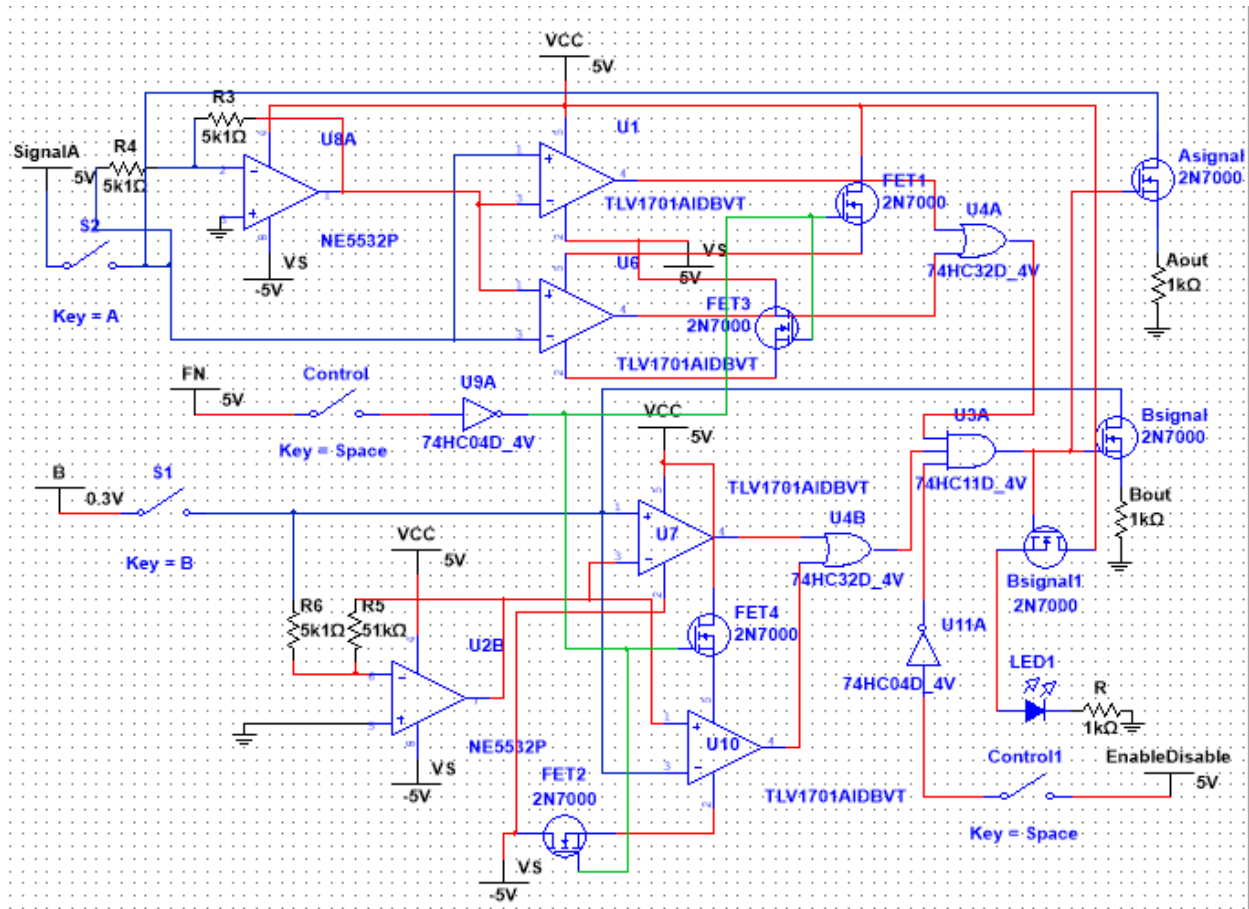


In the second case there are an odd amount of inputs, so connect the LED from the first one to the B input pin.

This is adding complexity, but my model is basically a logic to synchronize the signals. I can make more devices like 3 input, 4 input, just like logic gates there are more than 2 input logic gates.

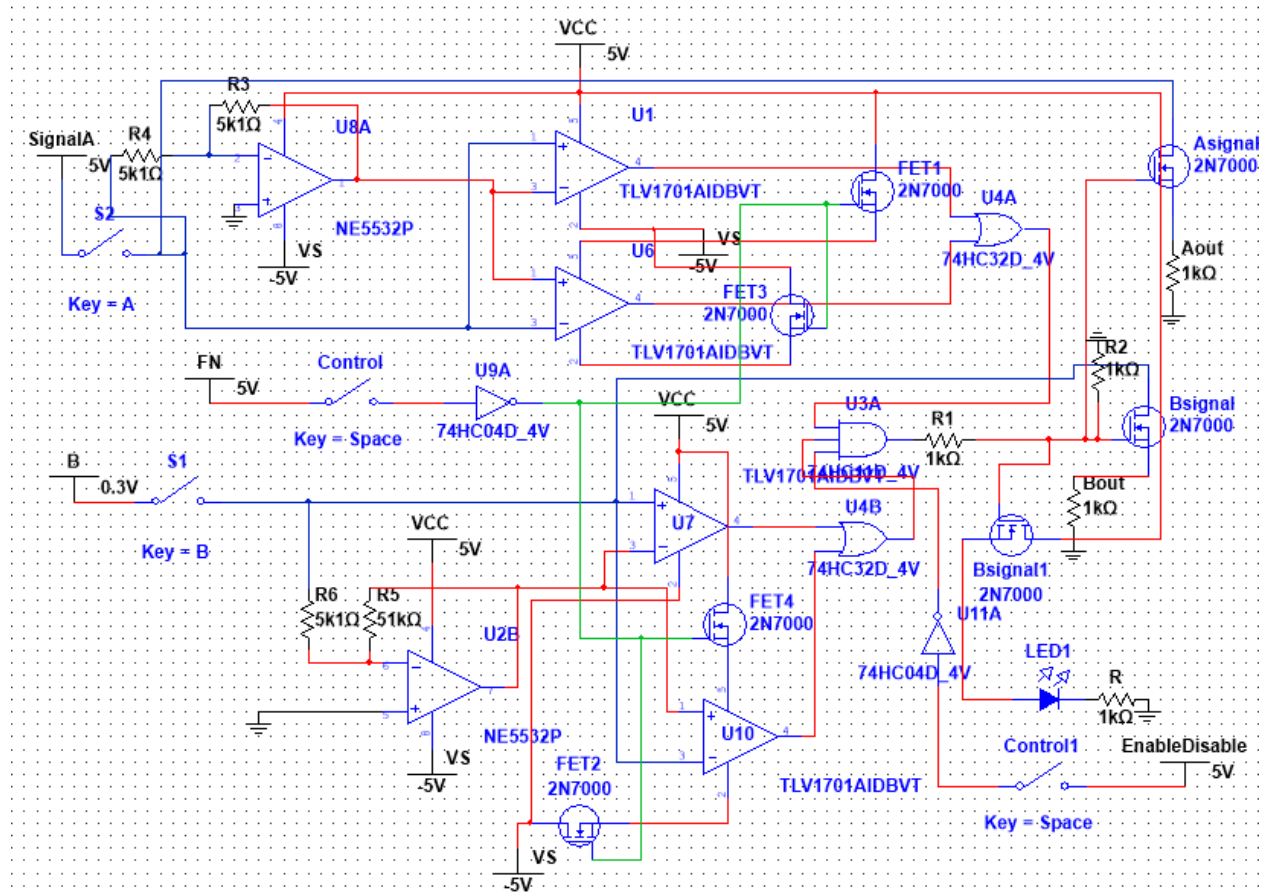
Below is the updated circuit.

Then, I figured out that tri states only allow digital signals. Through cascading, I think adding extra FETs outside after each output pin will be good.



11/28

I'm checking the datasheet for 2N7000, it says that the max V_{th} is 3V, min is 0.8 V. I will add resistors after the AND gate so that the output lies between 0.8V and 3V. The typical voltage is 2.1V, I can limit the voltage to around 2V.



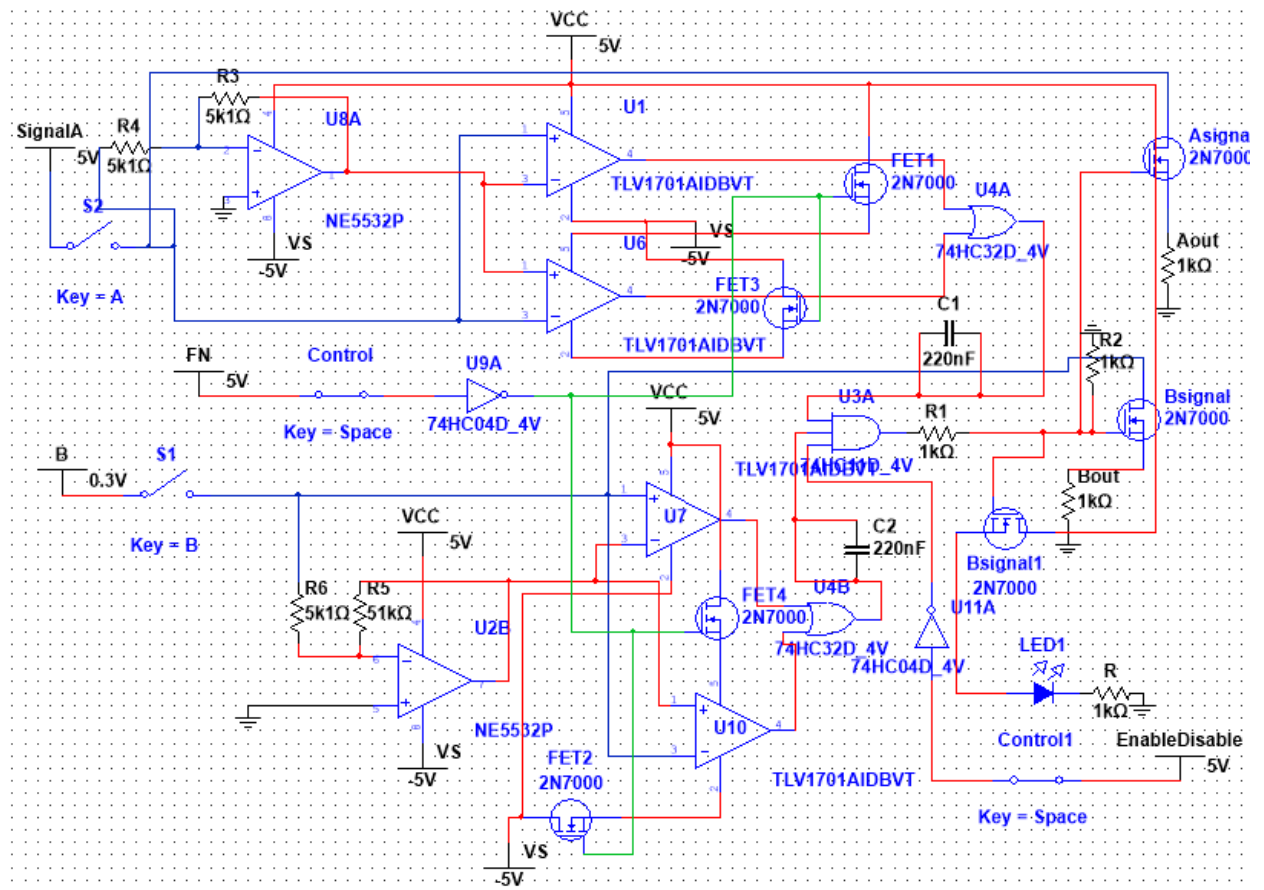
Updated!

11/30

1. I'm thinking that **if one of the input signals has certain 0s, like it's a sine wave, then at 0, my circuit will stop the signals from flowing in.**
2. I can **add a capacitor to extend the time of the circuit working a bit so that it doesn't need to turn off and on.** And the capacitor only maintains the circuit very shortly, short enough for the signal to come back. I add next to the end gate so that when one signal already comes, it will first charge the capacitor really fast, and if the signal becomes 0 or it disappears, the circuit maintains a a little longer and wait for the signal to cut. It will also prevent high peak voltage from turning on and off the MOSFETs, which prevents more energy loss. I need to consider if the capacitor delays

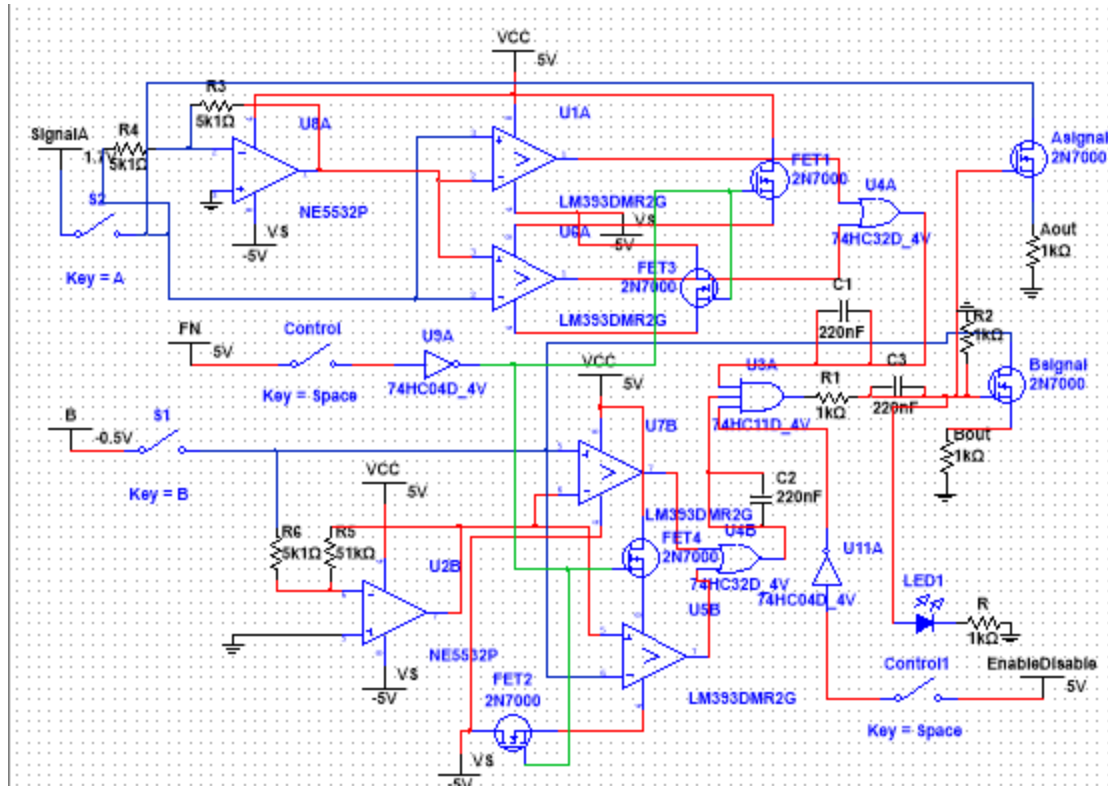
12/2

the signal when I build the circuit. This is the updated circuit.



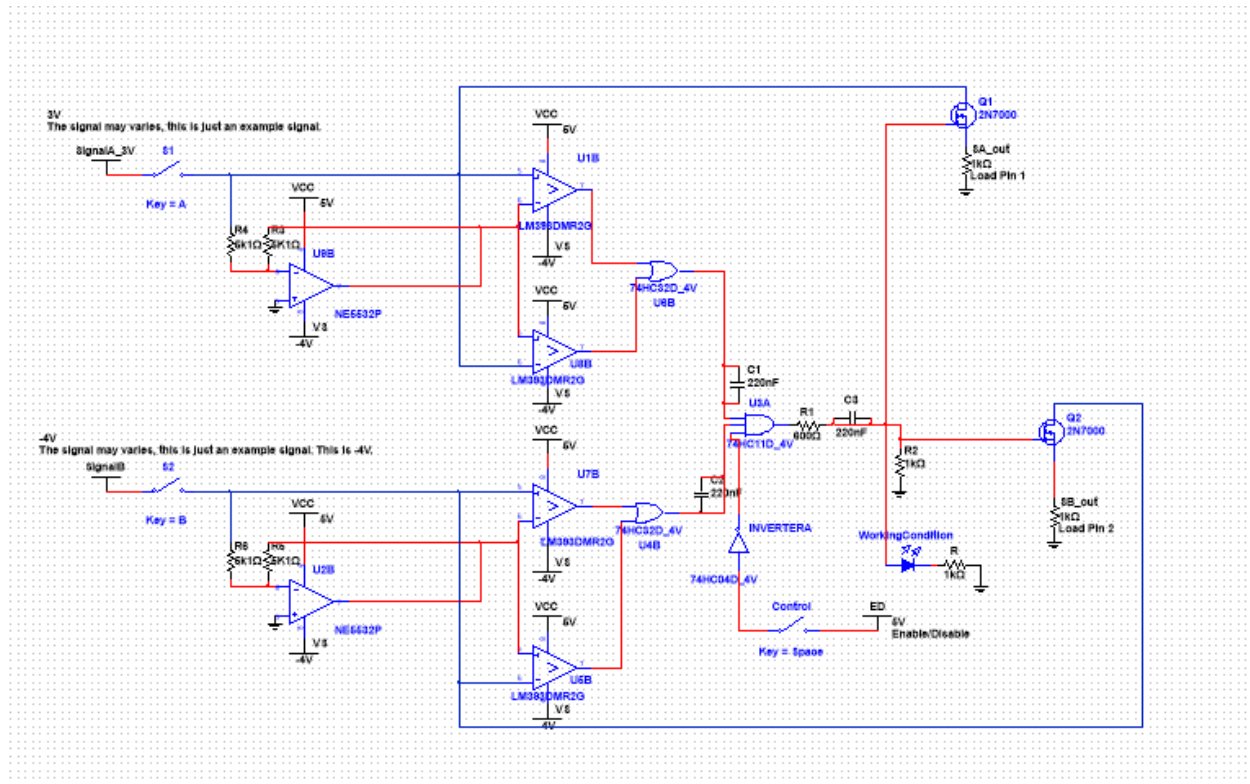
1. **I realized that LM393 works because according to the datasheet, I don't need to buy the TLV1704!**
2. **I asked Mr. Hanze to be my mentor, and he said yes.**
3. **I talked to my mentor today, he said my design is great, I can start building the thing.**
4. **I will also add a capacitor** after the AND gate to maintain the circuit working while the input is 0 at certain times.
5. **I also realized that I don't need an extra NMOSFET before the LED, the output voltage is already powering the LED.**
6. After changing components, Multisim now works. However, the output after the NMOSFET for A signal is now 1.69 V somehow, I realize that the VGS but not VG should $> V_{TH}$ (0.8V for 2N7000). I will find a way to make it work. I found that a Transmission Gate may work.
7. If I just use a multiplexer instead of a transmission gate, but there's no need, a transmission gate is simpler.
8. I will keep it simpler, set the output of the AND gate to 2.5V, the range of the signals accepted is then -0.5V to 1.7V.

This is the updated circuit.



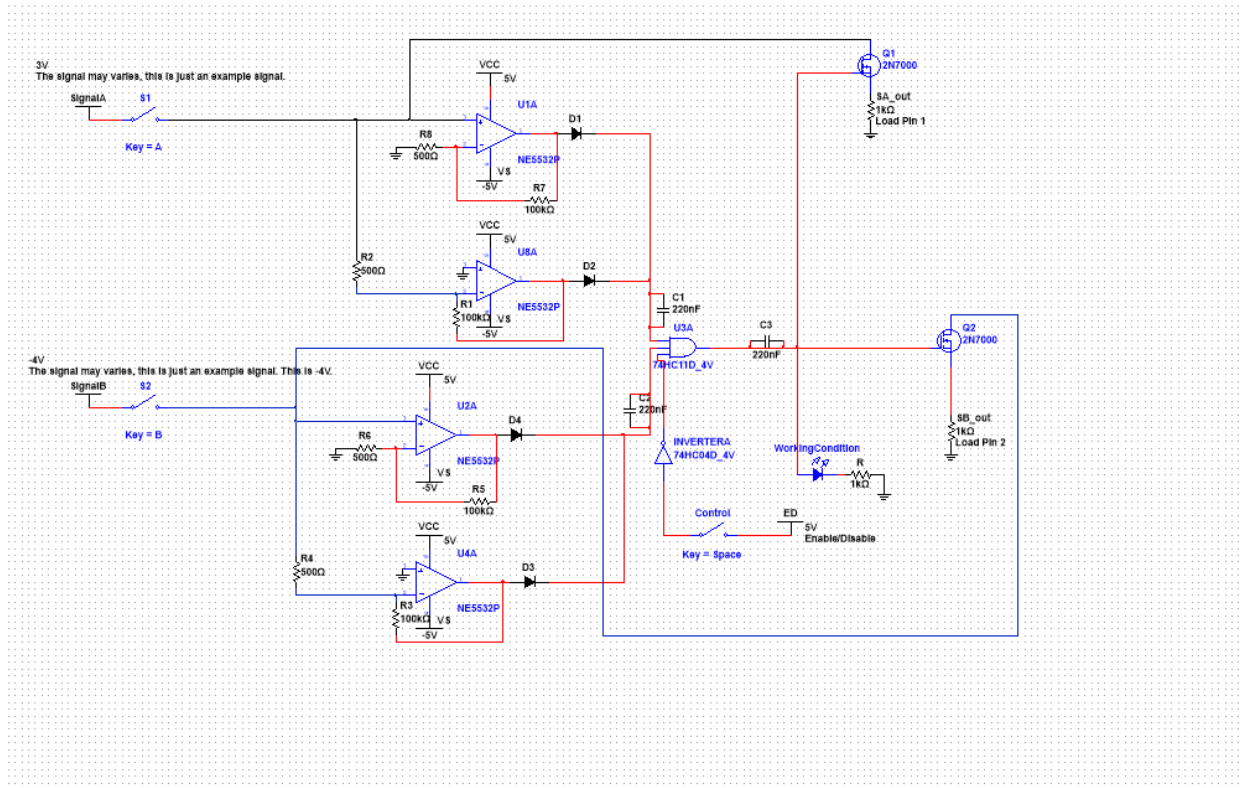
12/3

1. I made a mistake about VGS, it's the max and min VGS a FET can have, not the range. The problem is the one I found on 11/28.
2. I don't want the extra function to disable half of the comparators, there's no need. I'm also making the circuit cleaner.



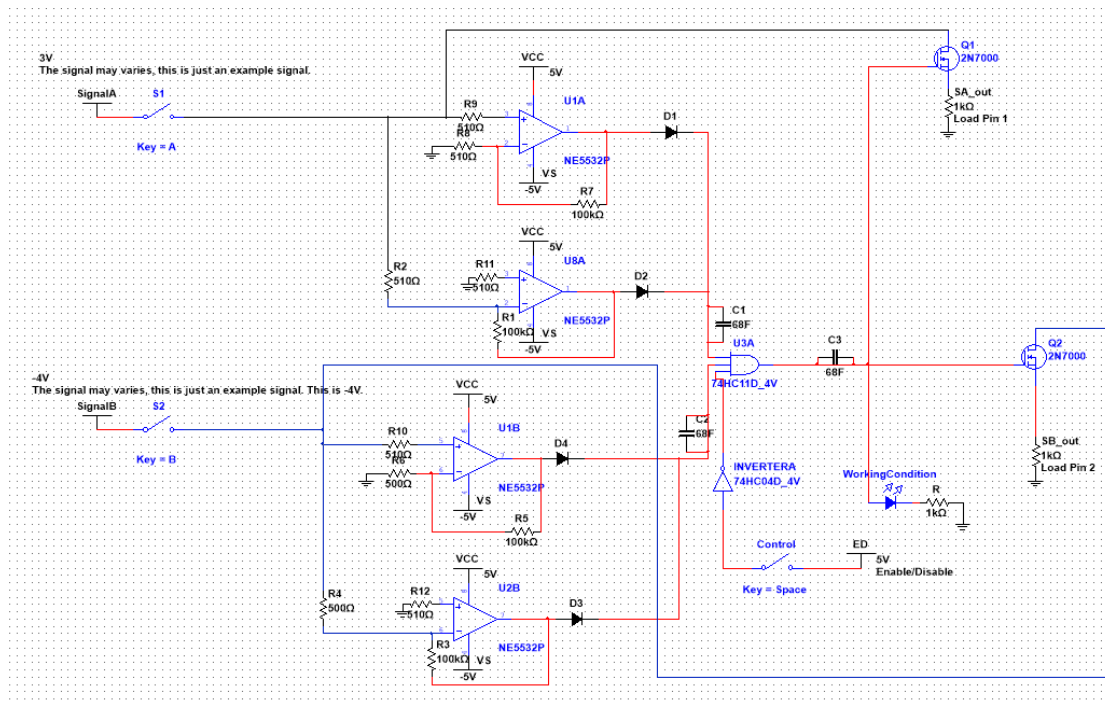
12/6

1. **Question: I realized that LM393 does not allow input to go below -0.3V, so I need to change a component.** (Before I wanted to use TLV1701, but I already ordered the logic gates on mouser but I did not order that because I made a mistake about LM393's minimum input).
Solution: I'm thinking if I use NE5532 (op amp) to substitute all the comparators, and have an extreme high coefficient like 100, then, if the signal arrives, one op amp outputs a high positive, one op amp outputs a high negative. Then, I will add a diode after the op amps so that only the high positive output goes to the OR gate, which prevents the OR gate from breaking.
2. I discussed with my dad about the situation, and we agreed to only use 2 op amps (1 negative coefficient, one positive coefficient) and connect a diode after both.
3. I also delete the OR gate.
4. I think op amp coefficient around ± 200 is enough since if it is too large, the offset voltage will be amplified to be assumed as signals.
5. This is the updated circuit.



12/8

1. I changed some values to the resistors and capacitors to the circuit. I add some resistors from the input pins to reduce noises.



2. I'm using 1N5819 diode, it's schottky diode so it has low VF.

3. I find out that diodes need current, but op amp outputs nearly 0A and only 5V in my case, so I will connect in series with a 10 Ohm resistor to create around 0.5A. But, the op amp can output current, and I'm using voltage for logic gates, so there's no need.

12/12

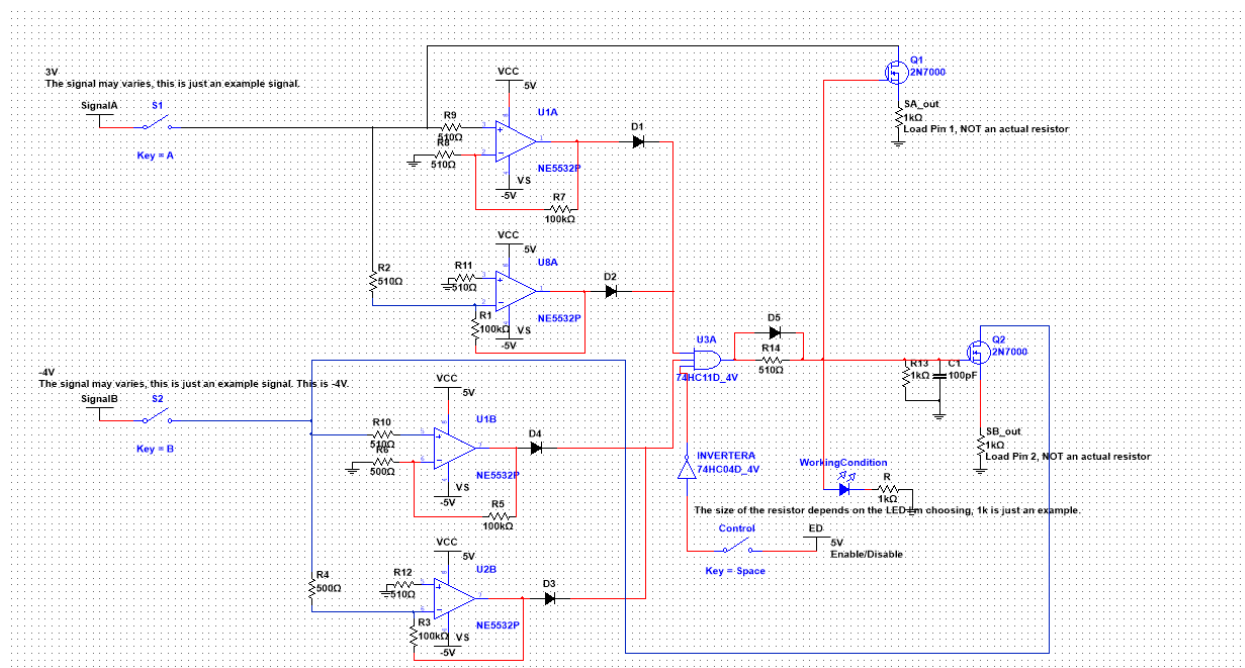
1. **If I use a MOSFET common gate amplification, connect output to 2 roads in parallel, 1 has 10 diodes to make the parallel circuit to around 7V, the other connects to the MOSFET which acts like a switch. This will reduce the cost even more, but require more testings.**
2. **But, my dad disapproves, MOSFET amplification is built for small signals.**

12/13

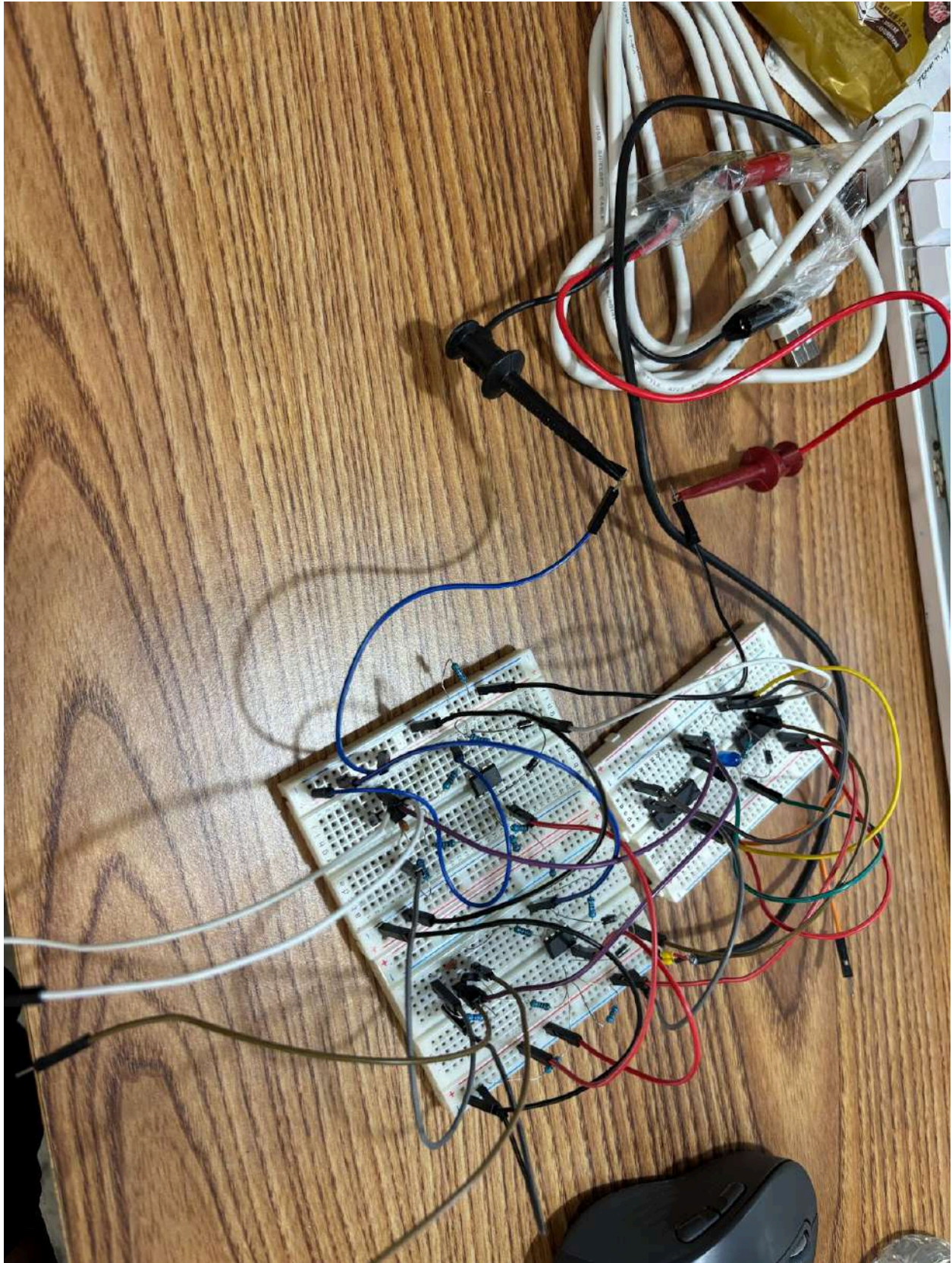
1. I want to add capacitors to the right value to achieve a delay of 100 ns to let the signals continue to pass through the nMOSFET even when they arrive at 0s. $t = -RC \ln(1 - V_{th}/V_s)$. I don't think I should add the capacitor outside 74HC11 because it may cause unwanted working conditions on the MOSFETs.
2. According to 74HC11's datasheet, I estimate V_{th} to be around 3.5V to maintain a V_{ih} . Input resistance is at least 5M (calculated by $5V / ((1 \times 10^{-6} A))$), the current is the max, which means it can go smaller, and the R can go greater). I will use a pF level capacitor then. Not decided yet.

12/15

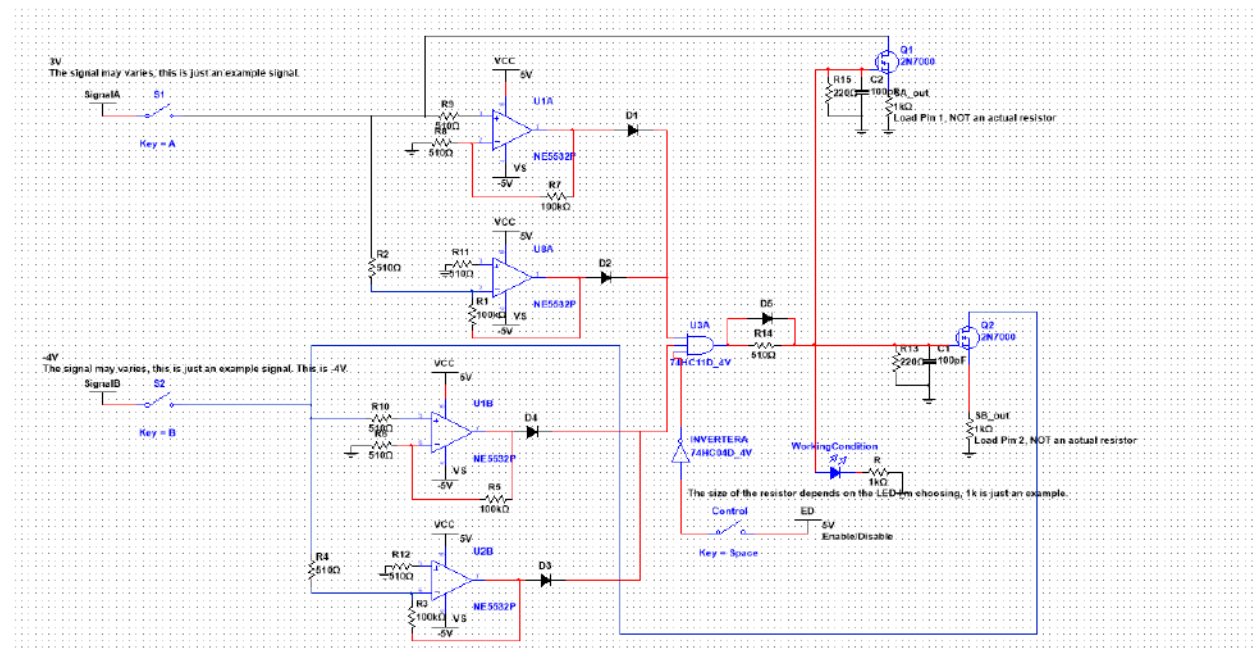
1. I changed some circuits to add a route for C_{ISS} to output voltage when the FET shut down, and I extend the time by RC, I add a 100 pF capacitor because the datasheet of 2N7000 does not have a typical C_{ISS}, so I will use 100 pF as a reference, and the output time can be longer based on C_{ISS}. I'm aiming for 100 ns turn off time. 5RC is approximately the time to discharge/charge. $5 \times R \times 100 \text{ pF} = 100 \text{ ns}$, $R = 200$.
2. I add a parallel circuit after the logic gate to achieve fast input, input will go through the diode part and go to the FET, and the resistance of that route is around 45 ohm from 74HC11. So the max turn on time is $5RC = 5 \times 45 \times 60 \text{ pF} = 13.5 \text{ ns}$. Then, the delay between signals should not exceed 13.5 ns, which is right in my expected range: <50ns differences.
3. **When the output of the AND gate is charging the capacitor, the timing is not affected by the resistor. However, when discharging it, the resistor forms a RC discharging loop, which extends the time.**



12/18:



1. Circuit completed without switches.



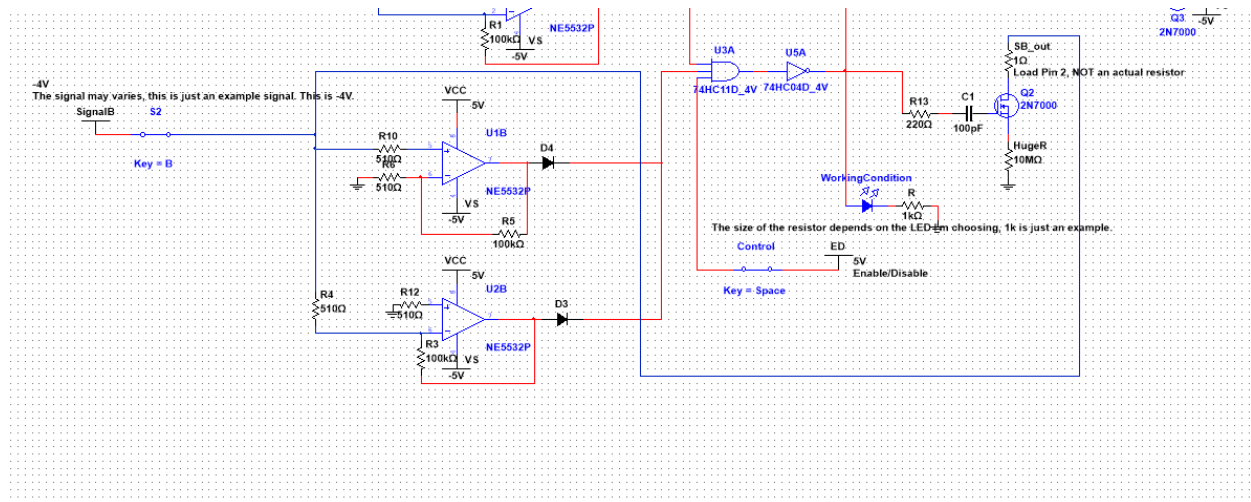
12/19:

1. One problem, when I connect all the signals to +5V, they are supposed to output +5V, and I connect an LED to show whether the signals arrive. However, the LED does not light up. It may be the LED is broken. I will use a multimeter to measure the output voltage. The output voltage somehow is around 2.2V when the input is around 4.9V. I test the outputs, they are around 4.2V, which are working as supposed to.
2. I get it, it may be because my signals are too high, and the fets are saturated, so it's not working as a switch but an amplifier, I will read more about MOSFET tomorrow and see if it works.

12/20

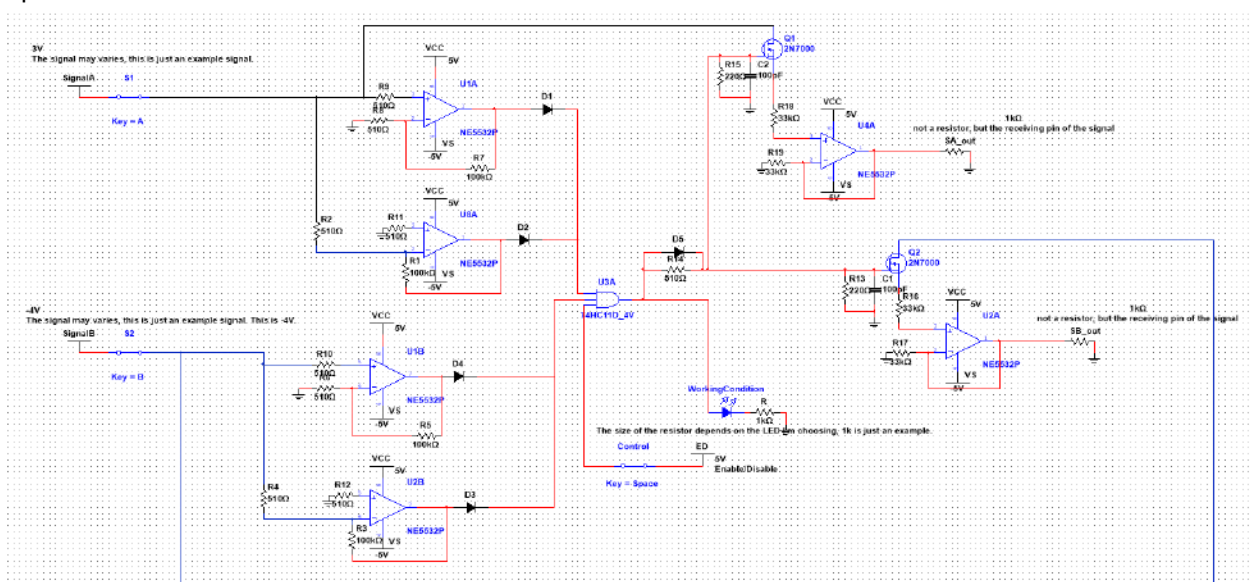
1. I tested the voltage of a variety of points with input signals of 2.5V. When I was measuring the Gate's voltage, it's around 1.7V. Then, I realize I connected the circuit wrong, now my LED showing working condition is no longer in parallel but in series with the Gates, I reconnected the circuit.
2. However, my output voltage of 74HC11 is around 1.6V, this is weird. I check the datasheet, it says it should output max VCC, but somehow it doesn't. I tested my VCC, it's around 4.9V. I tested 74HC11, it outputs around 4.8V when all inputs are high. Maybe it is because the 74HC04 or bad connection of the chip's pins to the breadboard, I disconnected one input pin, but the output is still high. I think this specific IC is not good. I changed an IC. It's the same. **I find what's wrong, my inverter gate is floating and connecting to nothing. I forgot to connect it to GND. I will not use the inverter gate because it will be overly complex, it adds an extra IC, I will let the user to input voltage to control it**
3. After I changed the component, the V_{oh} now is around 3.5V. However, the output signal is around 2.19V when the input is around 2.47V. This is fine because of the voltage drop of the FET. I will test more about my circuit and buy a CMOS switch. I'm thinking if I change it to a voltage follower it will also work.

4. When the signal is -2.518V, the output is the same.
5. I don't want to buy any more components, I'm thinking that if I change to using the NFET to connect to an extremely large resistor, and when the FET opens, all the voltage will go to the resistor but not the device. And if the FET does not open, all the voltage will go to the original device, so I will change the logic to reverse logic. I updated the circuit. I tested the Inverter, it requires low input to be connected to actual voltage but not nothing to reverse. Then, I need to change the capacitors to series to make the NFET open slower to achieve the goal of letting signals pass through even when zeros occur.

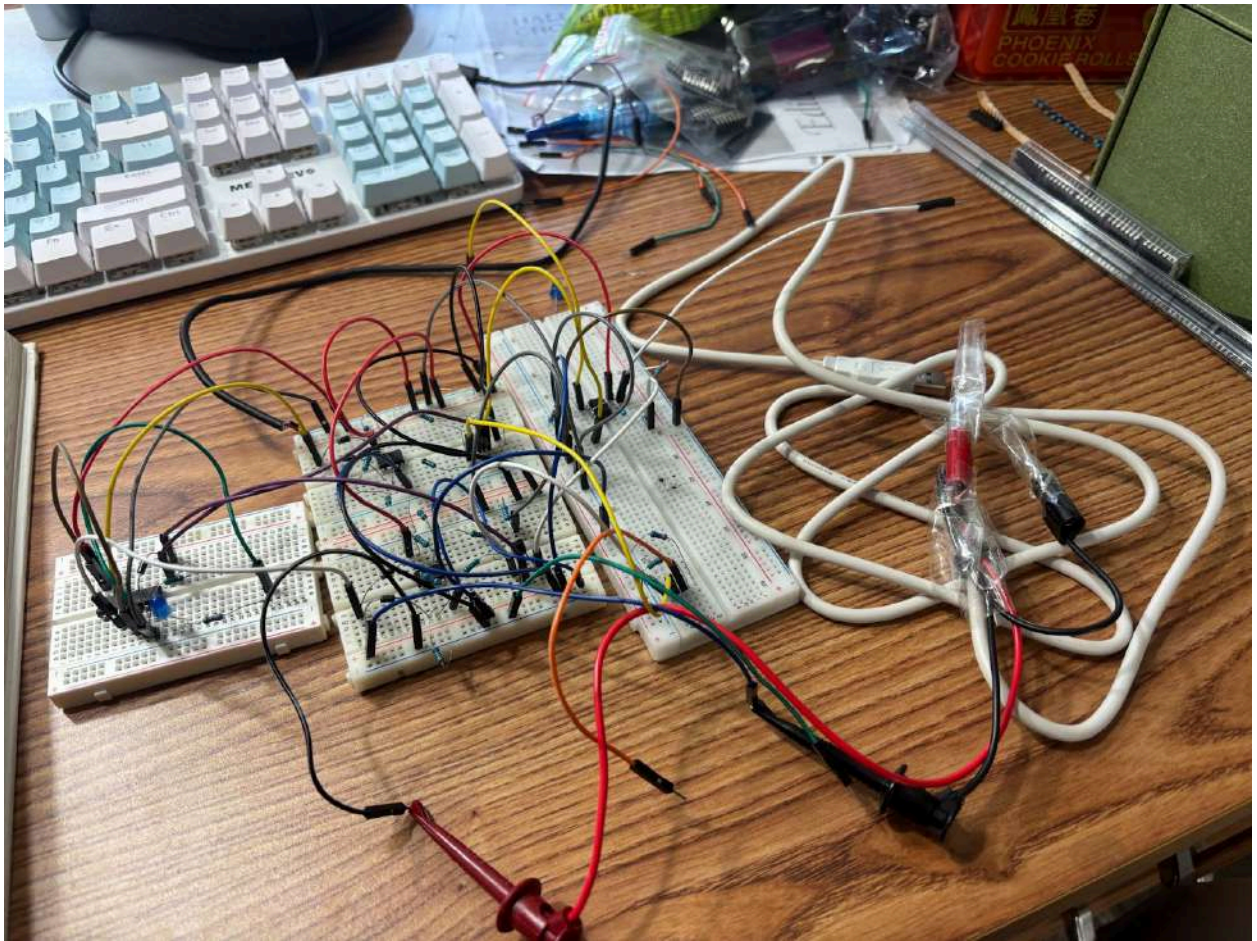


12/24

1. My dad asked me to try to solve the voltage drop by myself.
2. I'm thinking of adding a voltage follower after the NFET with large resistors to limit the current so that the voltage drop is almost 0 on the FET. I made a stupid mistake on making the receiving signal device in series with the FET, I cannot dismantle the device and make it in series.
3. Updated Schematic



4. Updated physical circuit.



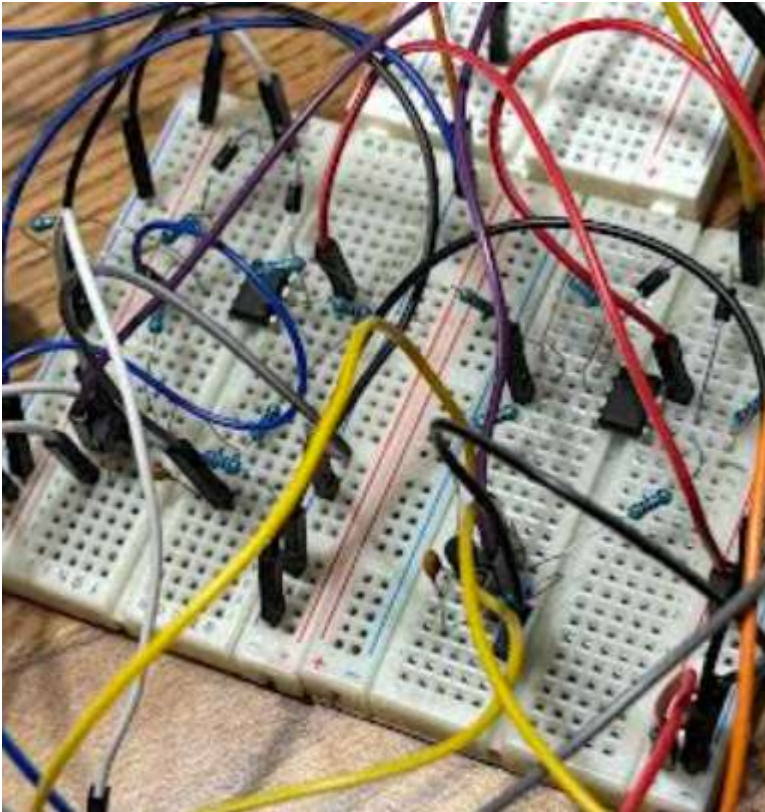
12/26

1. Vol of the logic gate is high, maybe the part I'm using is bad, I rewire and use another chip. The output voltage for the positive input is not good, it has now 0.6V voltage drop.
2. Maybe I connected the NFET the wrong way, D should be the input, S should be the output. Now I think I did the opposite. I figured out that the input now is too high, I connect to almost 5v, the FET is now saturated, it's an amplifier now. I checked my wiring, my NFET is connected correctly, so it must be at saturation.
3. Using the multimeter, the V_{in} and V_{out} are approximately the same.
4. I calculated my RC timing to discharge to 3V (max V_{th}), $t = RC \ln(V_{cc}/V_{th}) = 220 * 100p * \ln(5/3) = 11ns$. Discharge to 0.8V (min V_{th}), $t = 220 * 100p * \ln(5/0.8) = 40ns$. I will change to a larger C because the C of the FET is not guaranteed. So I want to minimize differences. At max V_{th} , I want at least 100ns. $100ns = 220 * C * \ln(5/3)$, $C = 890pF$. I don't have the exact size, I will use 1nF.

12/27

1. **My dad gave me an oscilloscope and asked me to figure out how to use it by myself. He also asked me to try to generate different signals by myself.**
2. **I downloaded the software and figured out that it can generate waves.**

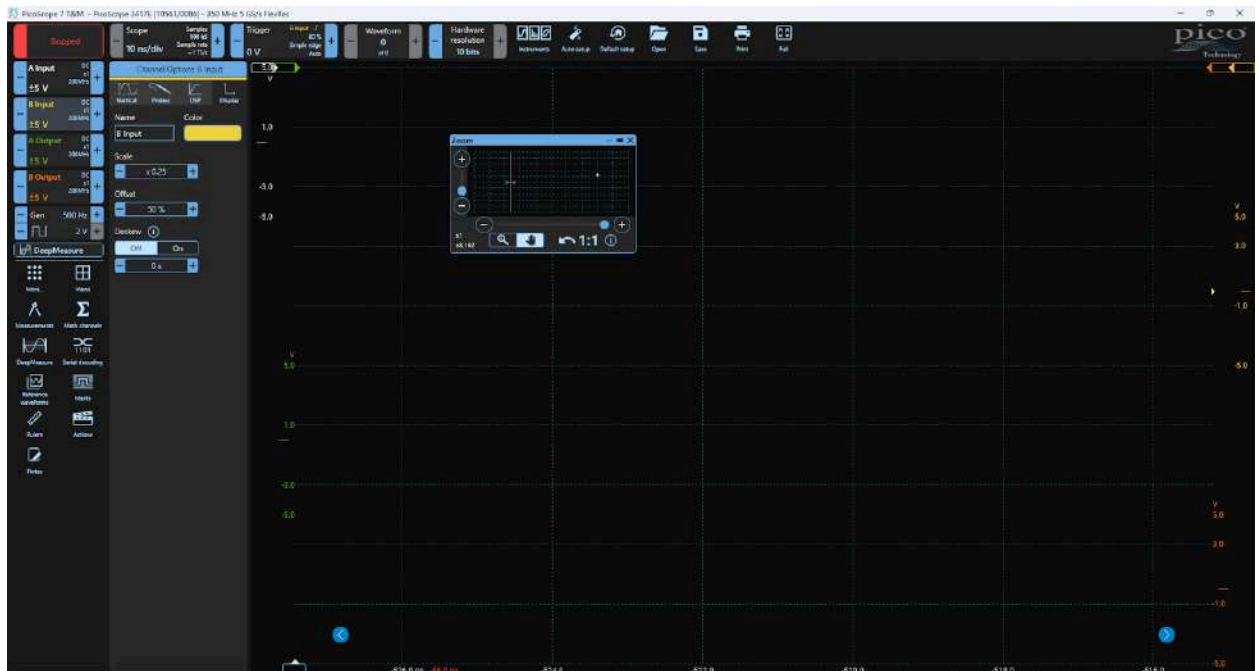
3. I updated the circuit. This is the changed part.



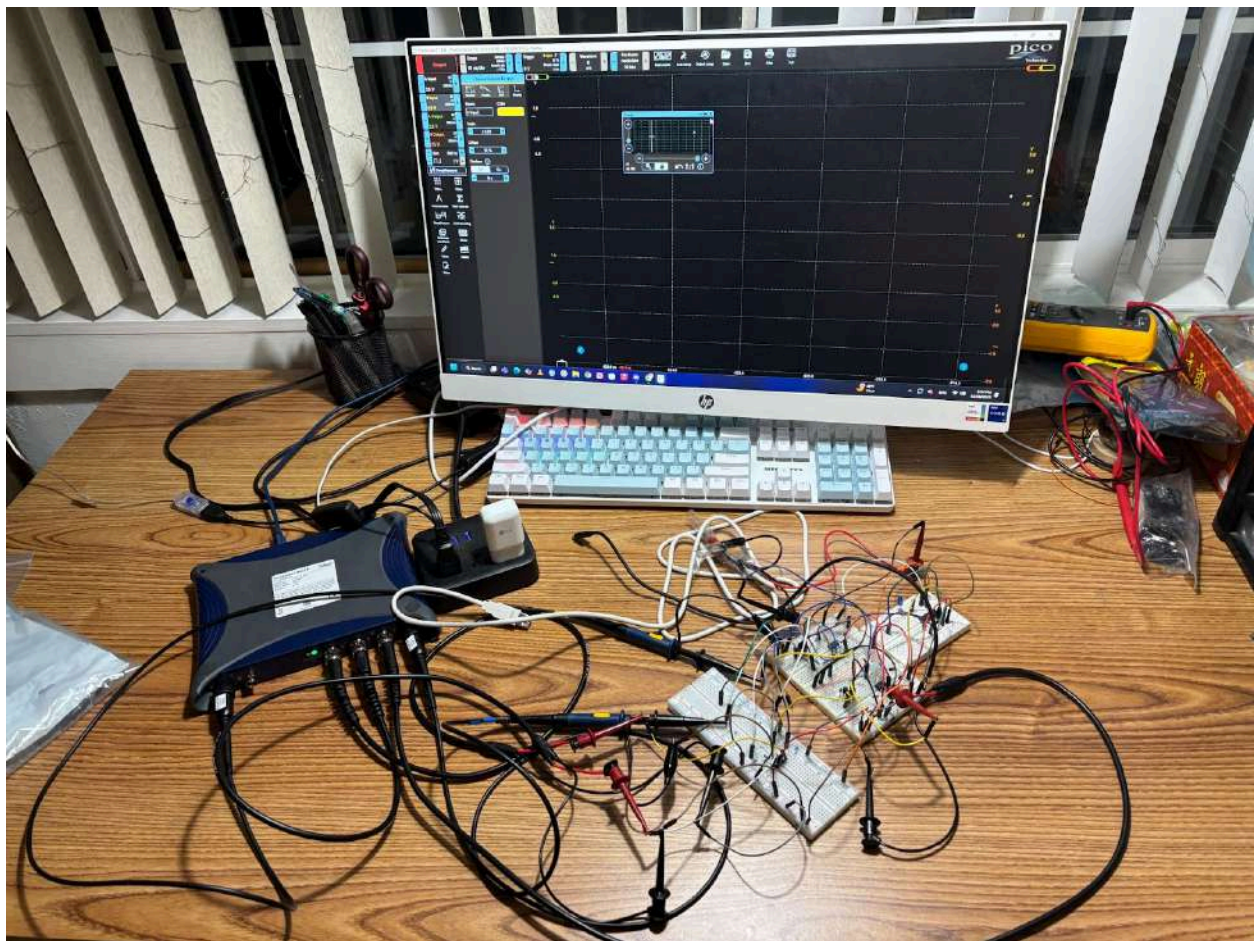
12/28

1. Physical circuit updated with new capacitor and connection.
2. Still learning how to use the oscilloscope, one of the wave generators is not working, I will do some research.
3. **I find it only has one built-in generator. I will try the same input waves for now.**
4. **I will put the signal source on one part of the board and connect the input signals at different times.**

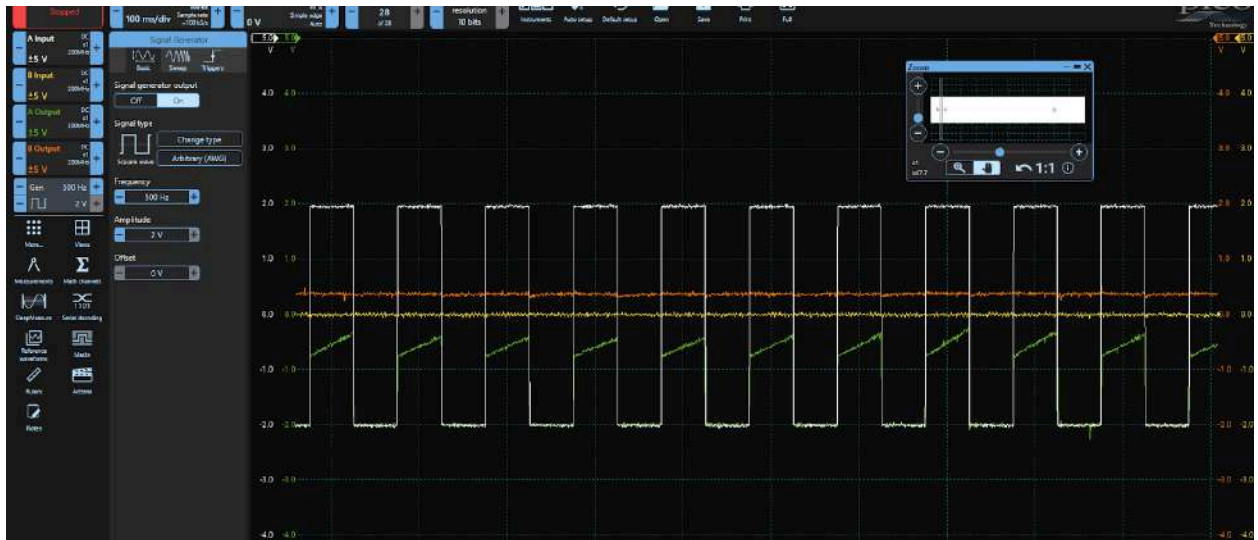
5. This is my screenshot of the working condition of the oscilloscope.



6. This is the updated circuit with the oscilloscope.



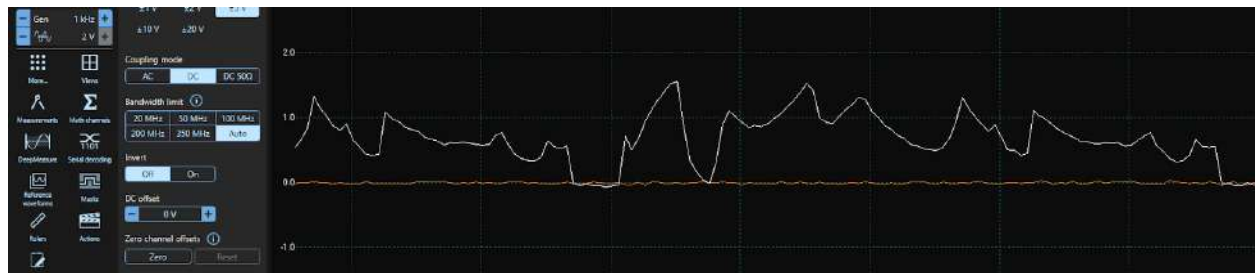
7. Somehow it is not showing the wave before AB both receive signals.



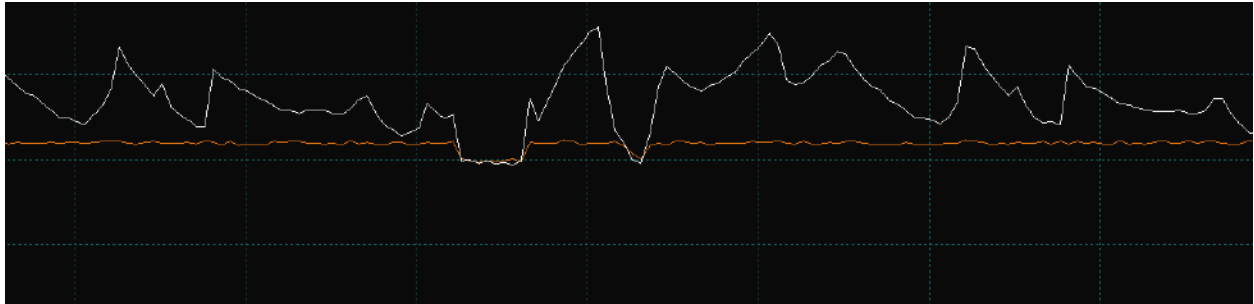
8. Only one signal comes, the output of the signal is not good at all. I talked with my dad, he asked me to try to test different points and see what happened. I tested various points, the Gate voltage is around 0.1, which is fine. I think maybe it's because I connected the NFET wrongly. And the output for the other signal is higher than 0, I think yes definitely I connect the NFET wrongly.
9. I tested the NFET's pins, and I connected them correctly. But, I found that I connected a capacitor incorrectly, the input pin did not connect to the gate for signal A.
10. I tested the output after the FET, it's not fine.
11. Maybe I accidentally put some other transistor rather than 2N7000 on the board.
12. I checked, I did use the correct NFET.
13. Maybe it's because the FET is not a good FET, it's broken.
14. The data is still the same.
15. I'm checking what happens if the other signal arrives first.
16. It's the same.
17. Maybe the NFET is not working at the condition I wanted to, now I'm doubting whether it is an amplifier or a switch now, I will check more tomorrow.

12/29

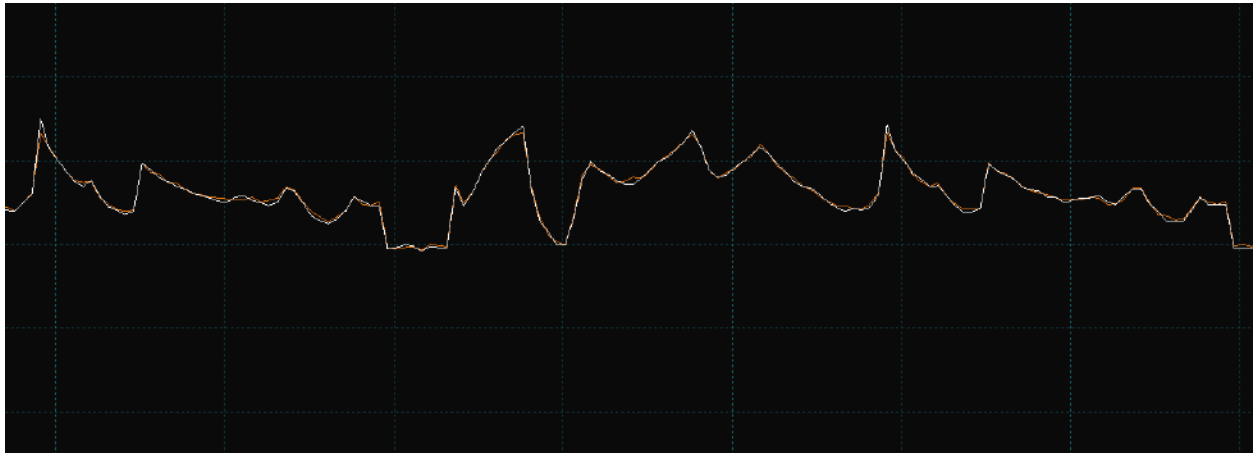
1. Testing the NFET: Gate voltage <1.6V



Gate voltage = 1.6V, the NFET is starting to turn on.



VG=3V, fully on.



Testing at different frequencies.

100kHz: very good.





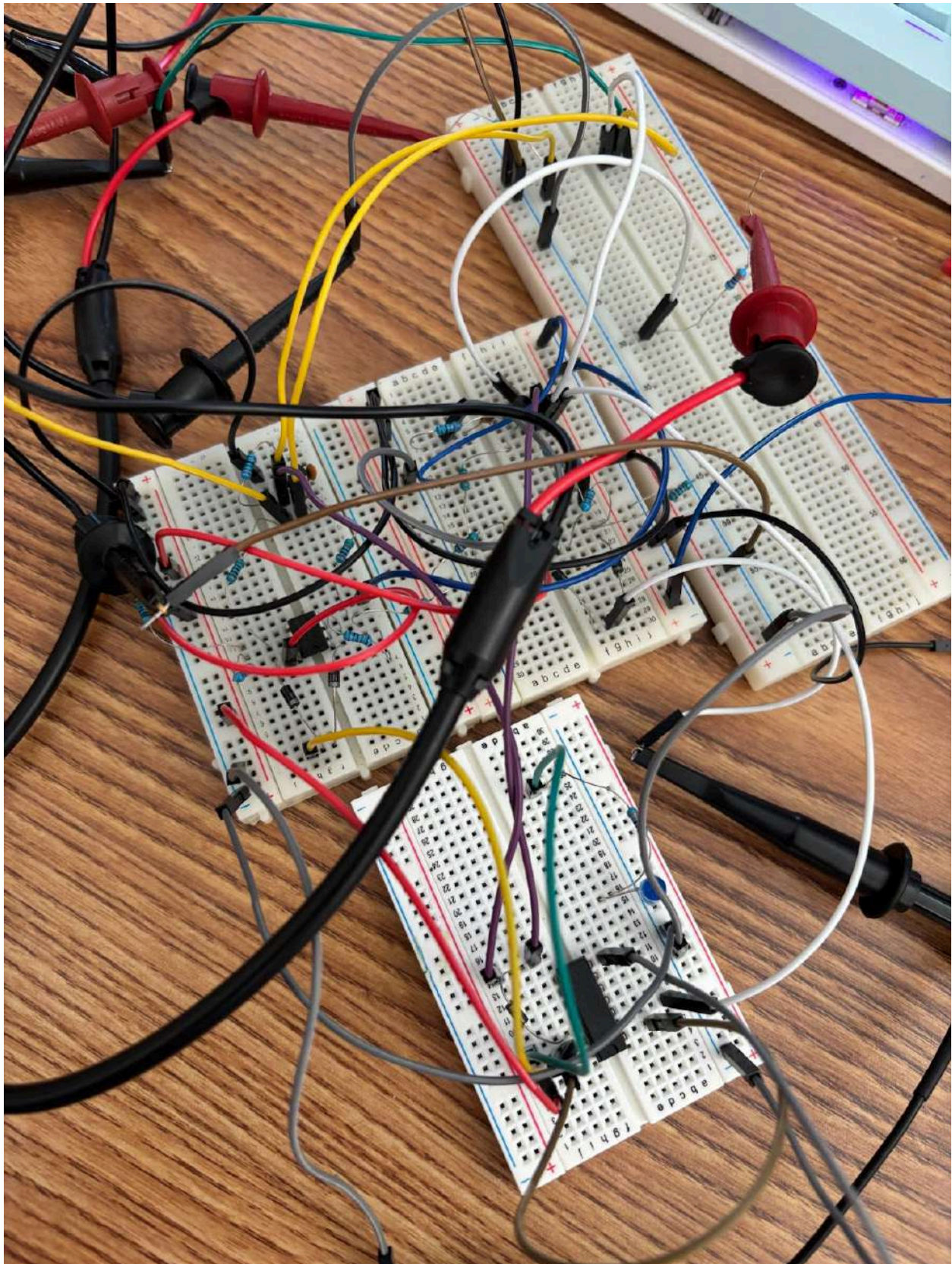
It works!

I change the input signal to be continuous, and it's working.

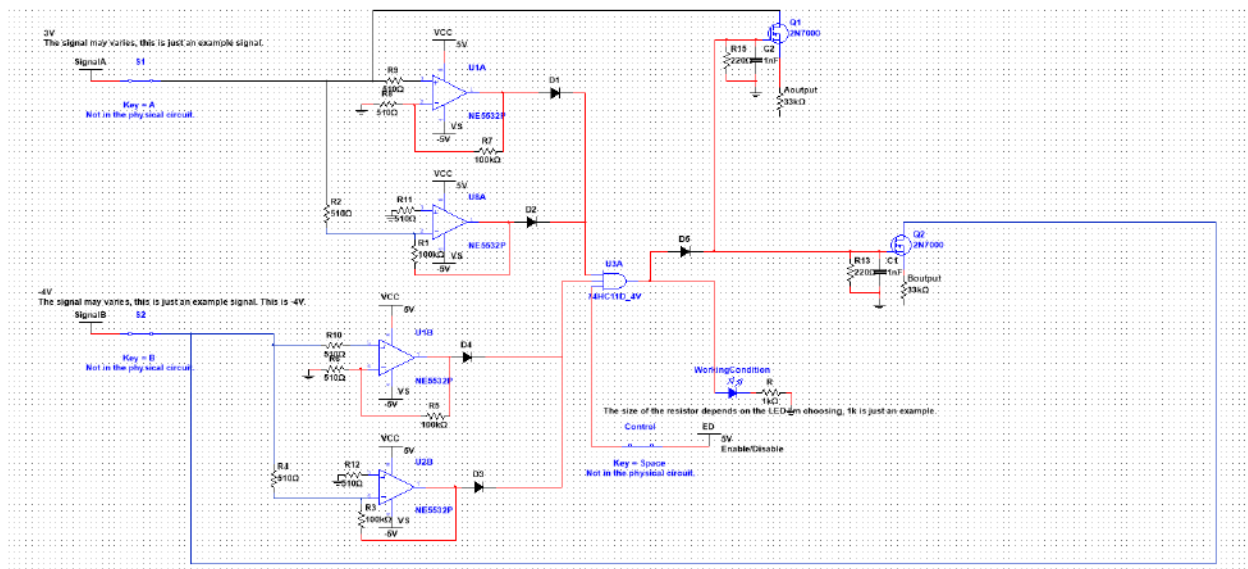


3. My dad questioned my resistor's function in parallel with the diode, he asked me to think about it by myself. I realized that the diode is already blocking the path back. So, I deleted it.

Physical updated.



Schematic updated.

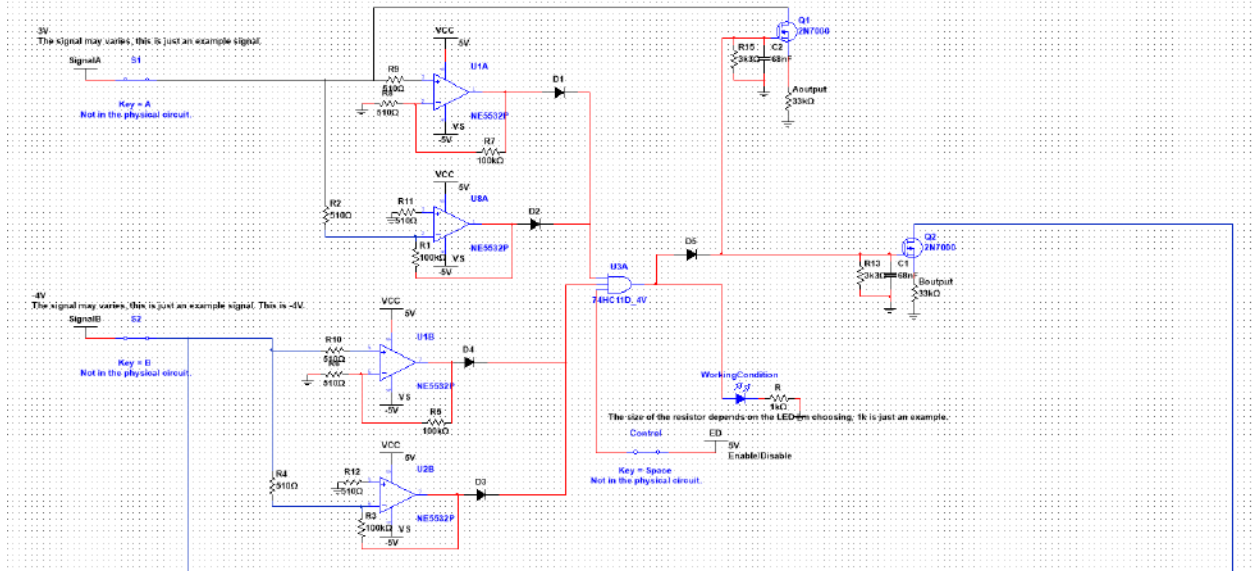
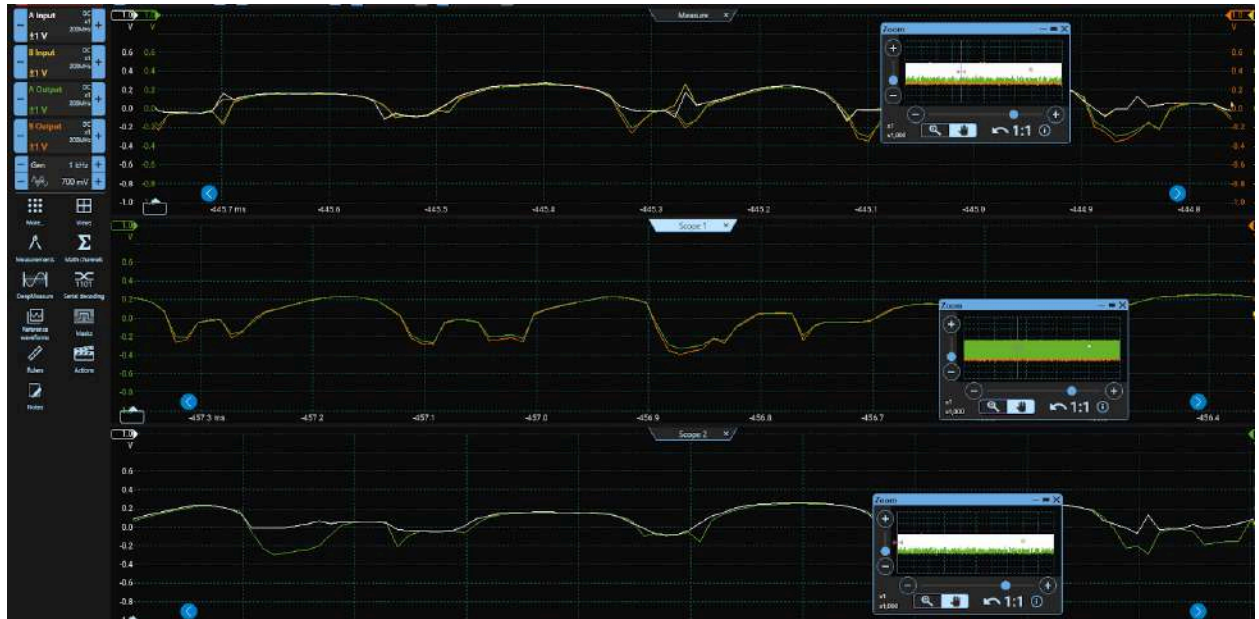


I want to find why with the voltage follower the circuit is not fine. I'm testing the voltage follower itself, but I did not find a reason yet. I did some research. It is likely because **NE5532 has an internal protection diode and if the voltage rises too fast, it outputs randomly. So, I will definitely not use a voltage follower.**

I have some other substitute op amp, but there's no need, it's adding complexity and noise. A 33k resistor solves the problem. My main worry was the voltage drop caused by MOSFET equivalent resistance, and if I use a large resistor like 33k, most of the signal will be at the 33k resistor.

12/30

1. Made minor changes to sizes of capacitors and resistors because the output voltage graph is not very good with a smaller RC time around 0V.



2. After the change, the wave is much better.

1/3/2026

1. I added a switch before the signals because previously I only used wires to connect and disconnect, but the wire is too sensitive and it creates noise when I touch the wire.

Data Collection begins today!

A=1V:



$A=0.5V$



$A=0.4V$:



Frequency Test: 30kHz:



20kHz:



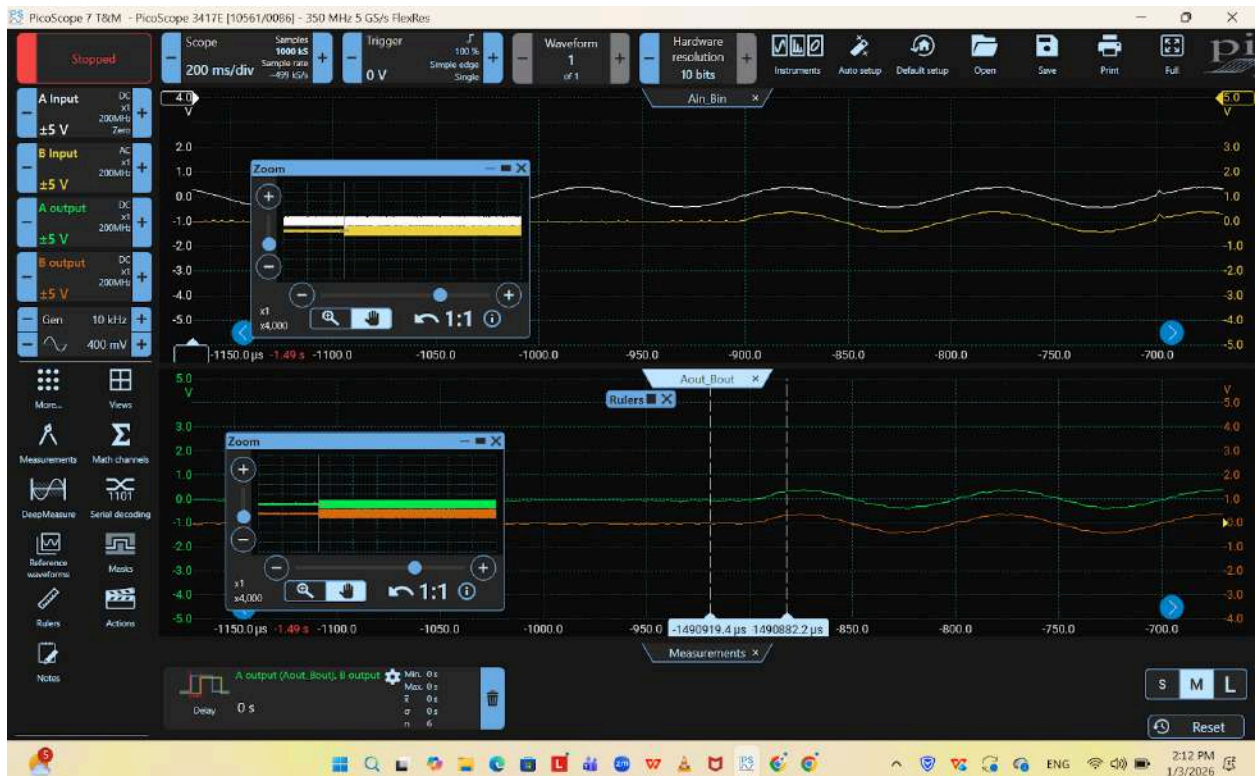
10kHz:



A first then B at 10k Hz sine wave Amplitude=400mV offset=0:



1.



2.

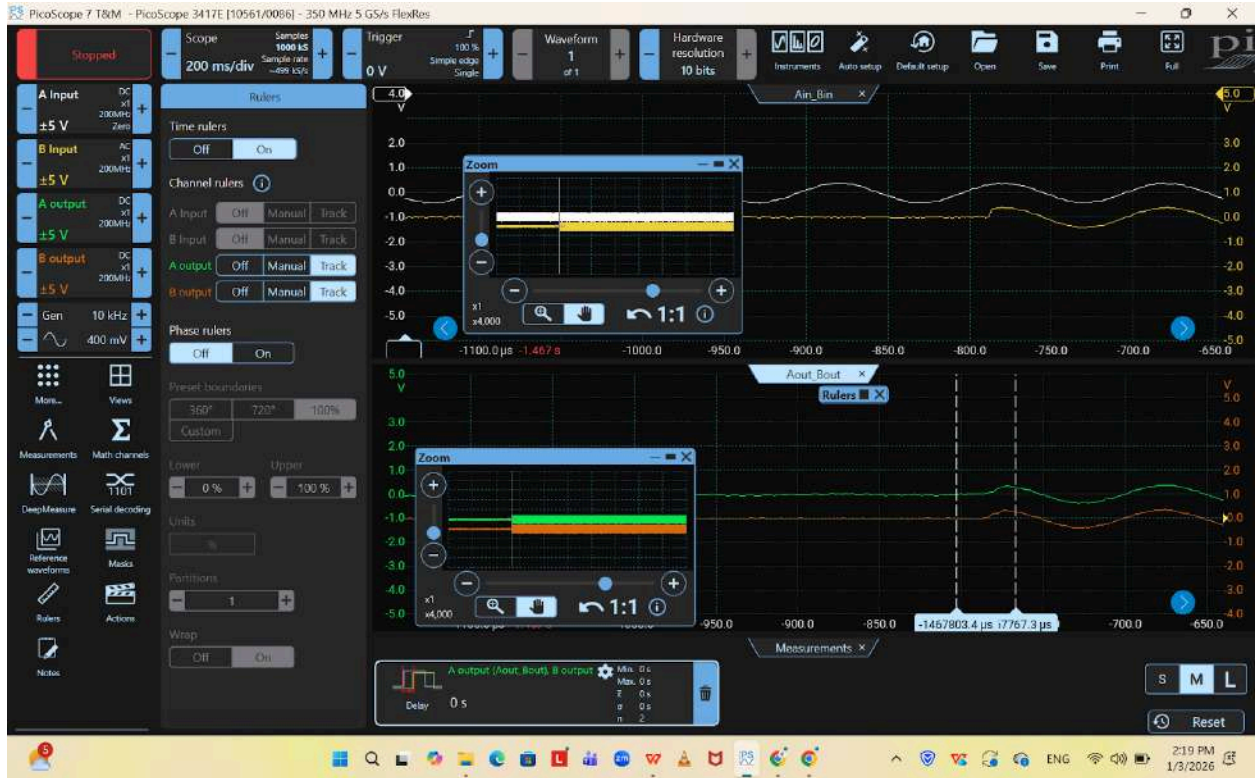


3.



4.

(1.907us)



5.



6.

7.

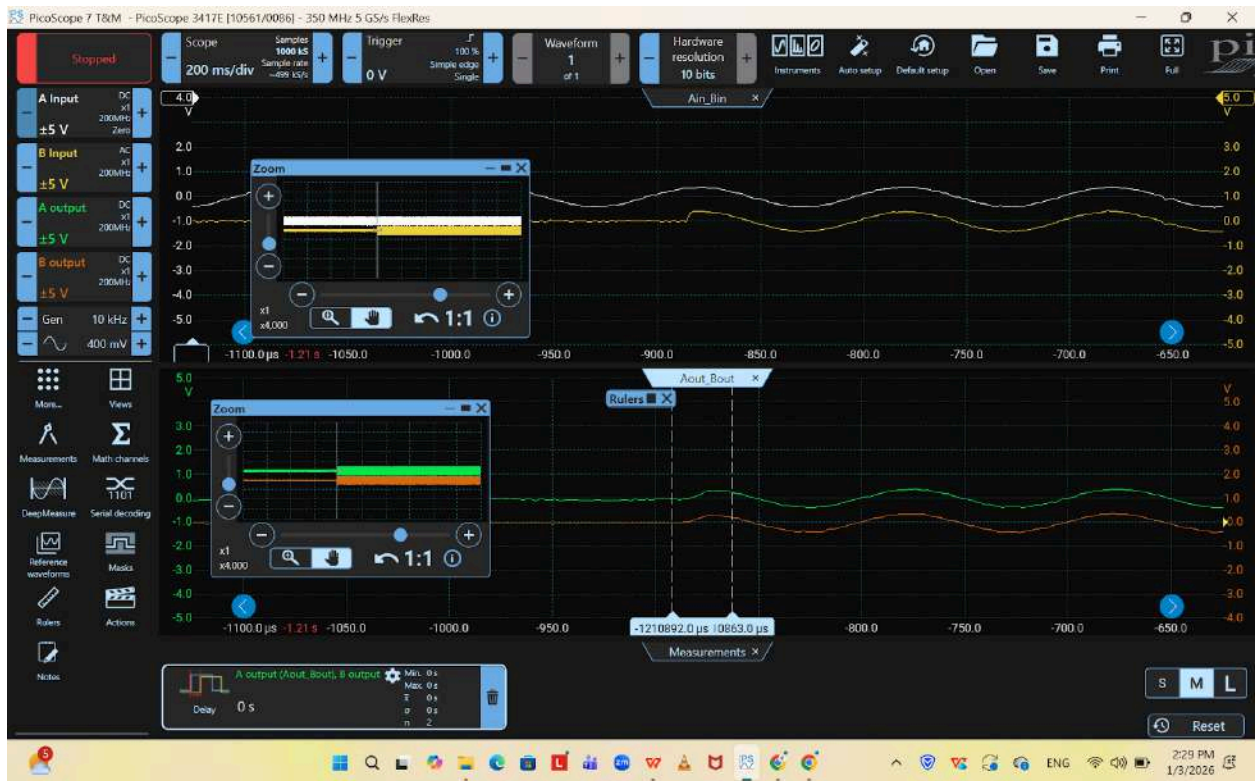


8.





9.



10.

11.



12.





13.



14.



15.



16.



17.



18.



19.



20.

B first then A at 10k Hz sine wave Amplitude=400mV offset=0:



1.



2.





5.



6.



7.

(2.027us)



8.





11.



12.



13.



14.



15.



16.



(1.013 us)



19.



20.

Square Wave Max Frequency Test:
3kHz(acceptable quality):



4kHz (bad quality):



A first then B at 1k Hz square wave Amplitude=400mV offset=0:



1.



2.



3.



4.



5.



6.



7.



8.



9.



10.



11.



12.





15.



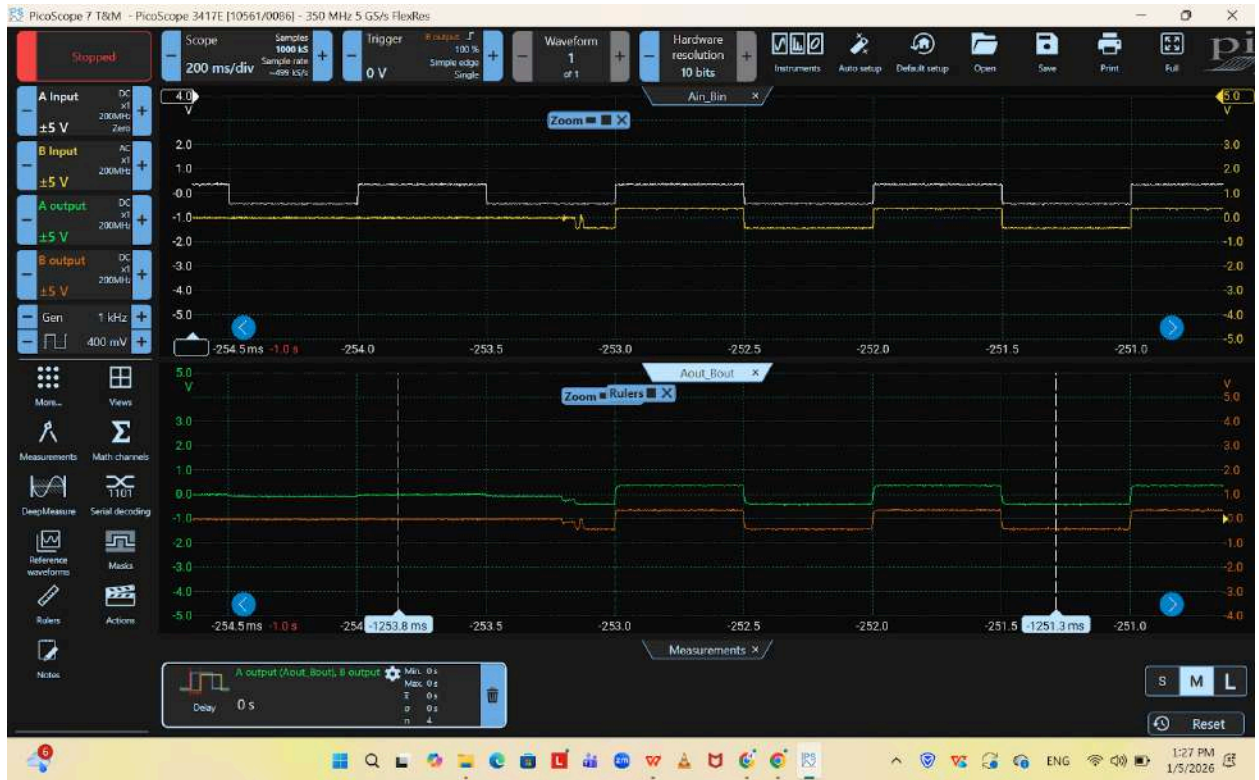
16.



17.



18.



B first then A at 1k Hz square wave Amplitude=400mV offset=0:



1.



2.







7.



8.





11.



12.



13.



14.



15.



16.



17.



18.



A first then B at 3k Hz square wave Amplitude=400mV offset=0:





3.



4.



5.



6.



7.



8.



(983.5ns)



11.

1/6/2026:

Continue data collection.



12.



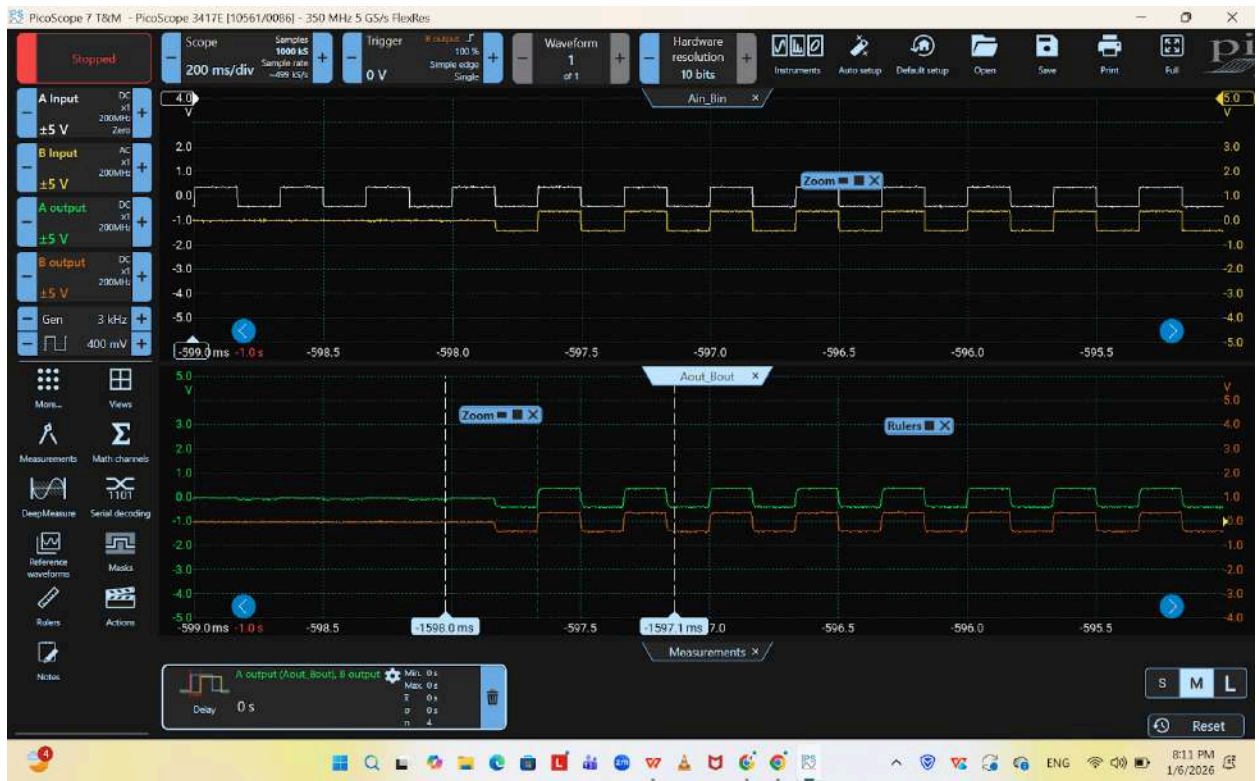
13.



14.



15.



16.



17.



18.



19.



20.

B first then A at 3k Hz square wave Amplitude=400mV offset=0:



1.



2.



3.



4.



5.
6.





7.



8.



9.



10.



11.



12.



13.



14.



15.



16.



17.



18.



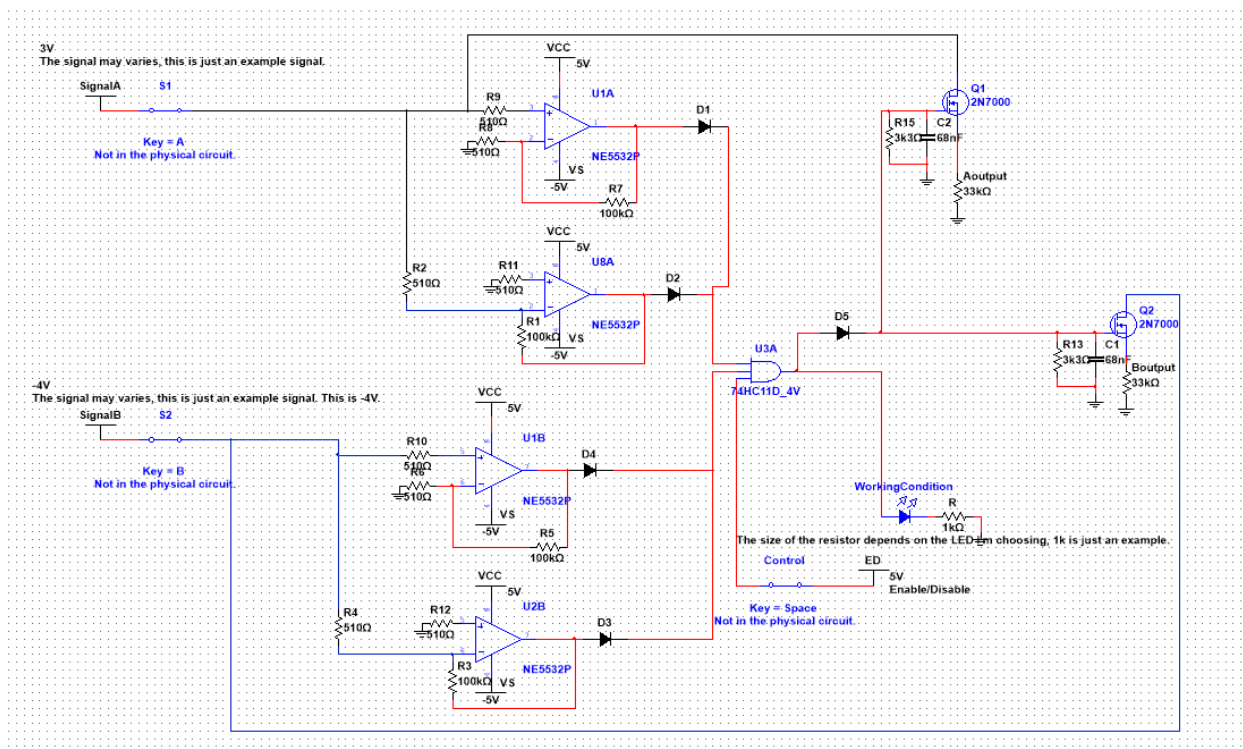
19.



20.

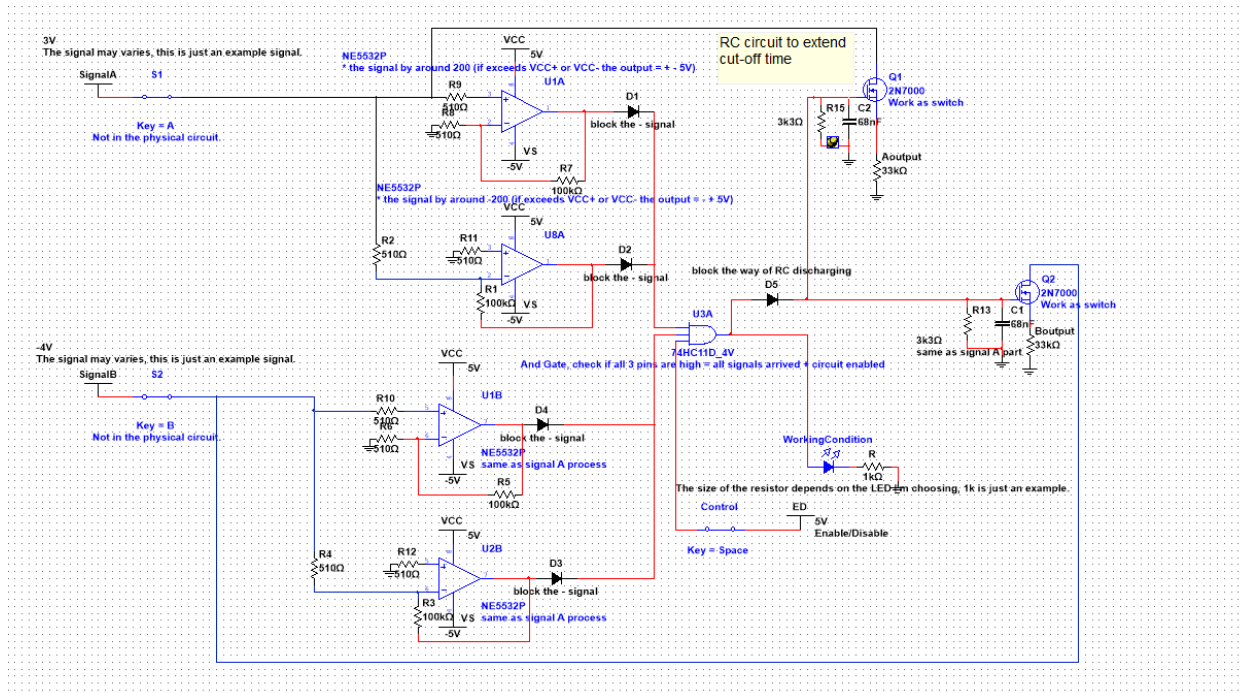
1/14/2026

1. Cleaned up my schematic.



1/15/2026

1. Updated circuits with comments.



2/8/2026:

1. Analyzed All data:



2/28:

I did a lot of research on the application of my prototype. I made these conclusions.

- Automotive (need precision of phase to 0.1-0.5 degree):** The **Steering Wheel** has an angle (where the driver wants to go). The **Electric Motor** has an angle (the position of the internal rotor). The **Resolver (generates 10kHz signals)** is a high-precision sensor. Tells the MCU which degree of the motor is at right now. It generates signals by interacting with magnetic fields. For the **Throttle system**, it manages how much air and power that the engine takes: a pedal uses 2 sensors (calculate how much the feet press the pedal down, it is redundant to make it safer.) and the MCU checks whether the phase of the 2 signals are valid at the same time. If they are not synchronized, the MCU may make wrong predictions. If the phases are modified, the MCU may also make wrong predictions. The MCU makes decisions to control the throttle body to control the output power of a car's engine.
- I read some source about ADCs and simultaneous sampling
- Industrial:** ADC synchronizes signals by the hardware itself, requiring a sync signal as a reference. However, if the ref signal is out of synchronization, it will need external synchronization. In sensors, there will be conversion from signals like vibration to analog signals/ They use multiple accelerometers to convert for IEPs.
- Medical:** ECG: 2 methods to sample ECG signals. 1. Simultaneous ADC capture signals. This introduces almost 0 delay. However, this requires using more ADCs or using a multi-channel ADC, which introduces hardware complexity and calculation cost. 2. Capture data sequentially. Use 1 multiplexer to let the ADC to capture signals sequentially. However, the multiplexer can introduce skew in timing and phase changes. (these can be solved through interpolation, but interpolation is only prediction, not reality). My method can be applied to **simultaneous ADC** so that it can capture all signals at the same time rather than continuing to capture invalid data at multiple channels. And, my method can let only when the device needs to perform calculation on the captured data, so it reduces the calculation cost of ability and lets the device rest when not all signals are valid.

3/1/2026

I calculated the price of my project by generating a BOM list. For certain parts, it has a minimum amount to buy that exceeds the need of my project's components amount. Still, the overall price is less than 5 dollars.

[illegible]

BOM							LCSC BOM Quotation												
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I find that the date of LCSC is in China, and it shows that it is tomorrow.

This is at mass production (producing 1000)

BOM										LCSC BOM Quotation									
BOM Data Export: 13/03/2025 12:40:31										BOM Export Date: 13/03/2025 12:40:31									
A07 B001a 0345486										Refresh Print Download									
The matching result is a real-time query result, availability and stock items are estimated only for your reference. Please refer to the official website for the latest information. All information is provided "as is" and is subject to change without notice. No liability is assumed for errors or omissions. No warranty is made without recourse to LCSC and its affiliates.																			
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183010001-15	1830-1111-1A-1	0805			1000	0.0077	0.0077	1830-1111-1A-1	1830-1111-1A-1	1000	0.0077	0.0077	0805	1000	In Stock		1830-1111-1A-1		
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18301000																			

_BOM [Export](#) [List Preferences](#)

EST. TOTAL **\$345.00** | \$0.35 each

Availability: ☒ In-Stock(7) ☒ Backordered(0) ☒ Discontinued(0) Quantity Multiplier = 1000

Uploaded Data	Product Detail	Match Status	Requested Qty X Multiplier	Order Qty	Stock Availability	Target Price & Lead Time	Unit Price	Est. Price	Offer Availability	Packaging options	Notes
1 Mfr.#: 743870WAG 54 Mfr. SER(嘉利) LCSC#: C5076372 Package: S00-321_U13-W13-L525-RD	Mfr.#: 743870WAG 54 Mfr.: SEP ICSC#: C5076372 2A 1 Independent ADZ 500 Impedance, 200k Ohm 325 Single Resistor, 0.8W Customer#:		5 x 1000 = 5000	5000	7300 in Stock LCSC: Backorder	\$ Target Price Lead time: days	\$0.007	\$3.50	LCSC: Backorder More Offer		
2 Mfr.#: F90002P00015 Mfr. FOIAN(富安) LCSC#: C0144032 Package: R0003	Mfr.#: F90002P00015 Mfr.: FOIAN ICSC#: C0144032 No Indicator 50V 100mA 25mA 425 0001 Chip Resistor - Surface Mount 0.8W Customer#:		0 x 1000 = 0	17000	47000 in Stock LCSC: Backorder	\$ Target Price Lead time: days	\$0.003	\$2.10	LCSC: Backorder More Offer		
3 Mfr.#: SNT4HC12R Mfr. TI(德州) LCSC#: C7043 Package: S0IC-14_U13-W13-P127-L50-D-BL	Mfr.#: SNT4HC12R Mfr.: Texas Instruments ICSC#: C7043 Customer#:		1 x 1000 = 1000	1000	17000 in Stock 12-17 business days	\$ Target Price Lead time: days	Quoted Discount \$0.0384	\$36.40	Other Suppliers See List More Offer	Full Page	
4 Mfr.#: 2N7002_R1_0000-CN Mfr. Onphipose(翊沛) LCSC#: C4237949 Package: S0T-23-2_U13-W13-P190-L524-BR	Mfr.#: 2N7002_R1_0000-CN Mfr.: Onphipose ICSC#: C4237949 N-Channel MOSFET 15mA 25mA 10V Surface Mount SOT-23 Customer#:		2 x 1000 = 2000	2000	2400 in Stock LCSC: Backorder	\$ Target Price Lead time: days	\$0.0076	\$4.40	LCSC: Backorder More Offer		
5 Mfr.#: C04040237R02M500 0T Mfr. HRC(宏发) LCSC#: C04040237R02M500 Package: C0402	Mfr.#: C04040237R02M500 0T Mfr.: HRC ICSC#: C04040237R02M500 0.4 x 0.25 50V Ceramic Capacitor X7R 5000 Customer#:		2 x 1000 = 2000	2000	27200 in Stock LCSC: Backorder	\$ Target Price Lead time: days	\$0.0064	\$0.60	LCSC: Backorder More Offer		
6 Mfr.#: SZY10-402B Mfr. YONGYU(永裕光电) LCSC#: C434447 Package: L000402-RD_BLUE	Mfr.#: SZY10-402B Mfr.: Yongyu Photoelectric ICSC#: C434447 Blue LED Indicator - Circular 0402 Customer#:		1 x 1000 = 1000	1000	996100 in Stock LCSC: Backorder	\$ Target Price Lead time: days	\$0.006	\$0.60	LCSC: Backorder More Offer		
7 Mfr.#: H058M Mfr. HOSB(A) LCSC#: C2987092 Package: DP-8_U13-W14-P154-L575-BL	Mfr.#: H058M Mfr.: HOSB(A) ICSC#: C2987092 Standard General Purpose Amplifier 2 Channel DP-8 Customer#:		2 x 1000 = 2000	2000	2340 in Stock LCSC: Backorder	\$ Target Price Lead time: days	\$0.0724	\$44.80	LCSC: Backorder More Offer		

This specification is for reference only. Please submit the BOM to request a formal quote. EST. TOTAL **\$345.00** | \$0.35 each

7 ITEMS SELECTED (In-Stock: 1, Backordered: 0)

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3/2/2026

Updated schematic appearance a bit.

